

# Does Per-Frequency-Band Graph Learning Help Copper Price Forecasting? A Rigorous Evaluation and a Diagnosis of Isotropic Embedding Collapse

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**Abstract**—We build a leakage-safe, capacity-matched protocol testing whether VMD-MFGNN, a per-frequency-band graph neural network, improves copper futures forecasting. Across eight models (seven baselines plus VMD-MFGNN), four horizons, and one 2010–2025 chronological split ( $N = 15$  input variables), none beats a zero-forecast baseline on mean absolute error, and VMD-MFGNN itself emits a single, constant-sign forecast at two of its four horizons, whose apparent directional accuracy there is a class-frequency artifact rather than measured skill. We diagnose why VMD-MFGNN’s per-band graphs show no advantage over a pooled alternative: graph embeddings in both the ablation and HPO-tuned models collapse to numerical zero ( $\text{RMS} \sim 10^{-11}$  against a Xavier-initialization reference of 0.2540), leaving every band’s adjacency exactly uniform. This collapse survives a mid-study fix to an unrelated graph-batching bug and is confirmed input-independent by a null-control experiment; a targeted fix (unit-normalizing the embeddings, excluding them from weight decay) does produce genuinely non-collapsed, differentiated adjacency structure, but the resulting model forecasts worse, not better, than the collapsed one past the first horizon — an intervention that cures the diagnosed pathology while hurting task performance, reported as an open, unresolved finding. Directional accuracy more broadly is uninterpretable without checking prediction sign concentration and a majority-class null alongside it: several models, including a tuned re-run that first appeared to reverse our negative finding, emit near-constant, single-sign forecasts whose directional accuracy exactly matches the test period’s class frequency for whichever sign they always predict, and we retract that apparent reversal once the artifact is identified. What we offer is an evaluation protocol rather than a winning architecture: a disclosed naive-baseline comparison absent from the directly comparable prior work we surveyed, a graph-collapse diagnosis confirmed independent of a separate implementation bug, and a warning about what directional accuracy alone can conceal, including in the flagship model this paper set out to propose.

**Index Terms**—copper price forecasting, graph neural networks, variational mode decomposition, multi-frequency analysis, commodity markets.

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## I. INTRODUCTION

Copper is among the most economically significant industrial metals, colloquially termed “Dr. Copper” in financial commentary for its perceived ability to signal macroeconomic

health. Global copper output reached approximately 22.8 million metric tonnes in 2024, with China consuming over 50% of global demand (IEA, 2024). The ongoing energy transition—electric vehicles require 53–83 kg of copper per unit versus 15–23 kg for internal combustion engines—is creating structural demand shifts that complicate traditional forecasting approaches (IEA, 2024). Forecasting errors translate directly into hedging and inventory decisions across the copper supply chain: mine operators and smelters price forward sales off short-horizon forecasts, while EV and grid-infrastructure manufacturers plan multi-year procurement off longer-horizon ones, so both the accuracy and the honesty of a forecasting method’s reported accuracy matter in a way that is specific to this market, not generic to commodities.

Recent advances in copper price forecasting have progressed along two largely overlapping tracks. The first track employs signal decomposition methods, particularly Variational Mode Decomposition (VMD), to separate copper price series into frequency components before forecasting each independently. VMD-LSTM (Liu et al., 2020) established the benchmark and is, as of this writing, the most-cited paper in this specific sub-area (290 citations per Google Scholar, accessed 2026-08), and secondary decomposition approaches (e.g., CEEM-DAN+VMD) represent the current frontier (Zhang et al., 2025a; Zhang and Ke, 2025). Most of these methods treat copper as an isolated univariate series, discarding the inter-commodity dependencies that drive prices, although a small number of very recent papers now combine VMD with graph neural networks (Section II).

The second track applies Graph Neural Networks (GNN) to model cross-asset relationships. GNNs have shown strong results in stock prediction (Cheng et al., 2022; Qian et al., 2024) and have recently been applied to copper via a Multi-view Graph Transformer (Sun et al., 2025), which constructs a single graph over undecomposed prices. Separately, at least one recent paper, MBTI-Net (Li and Liu, 2026), combines VMD decomposition with a GNN for copper futures, but pools the decomposed modes into a single graph rather than constructing one graph per frequency band. In both cases, a single graph structure collapses whatever multi-scale dependency structure exists across frequency bands into one static representation.

The premise motivating a per-band graph is intuitive and, on its face, plausible: commodity relationships are plausibly

multi-scale, so the copper-dollar correlation at low frequencies (reflecting long-term macroeconomic co-movement) may differ structurally from the copper-dollar relationship at high frequencies (reflecting daily trading noise), and a single graph, pooled or otherwise, cannot represent both simultaneously. **This paper does not assume that premise; it tests it.** We built VMD-MFGNN, a Multi-Frequency Graph Neural Network that constructs a *separate* learned inter-variable graph for each VMD frequency band, exactly the architectural distinction MBTI-Net-style pooling forgoes, and evaluate it against seven baselines and a matched pooled-graph ablation under a leakage-safe, capacity-disclosed protocol. We report what we found even where it complicates the paper’s own premise: none of the eight models tested, including VMD-MFGNN, beats a trivial zero-forecast baseline on mean absolute error at any horizon, and direct inspection of the trained model’s learned graphs shows the per-band mechanism did not learn differentiated relational structure at all—its embeddings collapsed to numerical zero during training, a failure mode we diagnose quantitatively in Section V-E1 and confirm survives independently of a separate graph-batching implementation bug found and fixed during this study. VMD-MFGNN’s own headline result is not exempt from this paper’s diagnostics either: it emits a single, constant-sign forecast at two of its four horizons, so part of what looks like its own directional performance is itself an artifact this paper catches and discloses (Section V-H). This reframes the paper’s contribution from an architectural proposal to an evaluation study with a diagnosed negative result, including about the very model the paper set out to propose.

VMD-MFGNN (Variational Mode Decomposition Multi-Frequency Graph Neural Network):

- 1) Decomposes a multivariate commodity/macro time series ( $N = 15$  variables, Section IV-A) into  $K = 5$  frequency bands via VMD, using a leakage-safe rolling window refit at every trading day;
- 2) Constructs a *separate* learned inter-variable graph at each frequency band, intended to capture scale-dependent cross-commodity dependencies;
- 3) Applies Graph Attention Networks (GAT) per frequency band for spatial modeling and an LSTM for temporal modeling;
- 4) Fuses band-level predictions via learned attention weights.

This paper is organized around four research questions, posed here before any results are reported so that Section V-I’s answers have a clear antecedent:

- **RQ0:** Does any model tested here, including VMD-MFGNN, beat a trivial zero-forecast (do-nothing) baseline?
- **RQ1:** Does constructing a separate learned graph per frequency band outperform pooling the bands into a single graph?
- **RQ2:** Does VMD decomposition, and does end-to-end learned (versus fixed correlation-based) adjacency, add measurable value over the alternatives?
- **RQ3:** Does VMD-MFGNN differ significantly, in the

Diebold-Mariano sense, from standard forecasting baselines?

Our contributions are:

- **A rigorous, leakage-safe, capacity-matched evaluation protocol:** a capacity-disclosed eight-variant ablation (Section IV-D), a Diebold-Mariano significance test with a multiple-testing correction (Section IV), and a naive random-walk baseline comparison we did not find reported in any of the eleven most directly comparable works in Table I. We build the protocol first and treat the architecture as the object under test because Kapoor and Narayanan (2022) document, across seventeen scientific fields, that leakage and evaluation shortcuts, not modelling choices, are the dominant driver of irreproducible ML-based science; this study’s own history bears that out on both counts (Section III, Section V-H).
- **A diagnosed, quantified mechanistic account of why the proposed graph-learning mechanism failed:** we trace the per-band adjacency’s collapse to isotropic norm decay in the graph embeddings, rule out the dead-ReLU explanation directly, and confirm the finding survives an independent graph-batching bug found and fixed mid-study (Section V-E1) — a citable finding for other low-node-count learned-graph architectures in financial GNN research, MBTI-Net-style designs included.
- **An honest reference point most comparison papers in this area omit:** every model, including the proposed one, is compared directly against a trivial zero-forecast baseline, and none beats it on MAE at any horizon (Section V).
- **A targeted, verified fix for the diagnosed collapse, and an honest account of its limits:** unit-normalizing the graph embeddings and excluding them from weight decay mechanically resolves the collapse (Section V-F) and, applied inside the real ablation run, produces genuinely differentiated structure (Section V-C) — yet the resulting model forecasts *worse* at every horizon past  $h = 1$ , a puzzle we report as open.
- **A general methodological finding about directional accuracy:** several models, including a re-run originally thought to reverse this paper’s central negative result, emit near-constant, single-sign forecasts whose apparent directional accuracy is a base-rate artifact rather than measured skill, caught only by checking prediction sign concentration and a majority-class null directly (Section V-H); we claim no priority for the underlying base-rate concern, which Cheung (2026) raises for equity forecasting, but add the sign-concentration and base-rate-identity machinery that detects a degenerate forecaster retroactively.

## II. RELATED WORK

### A. Copper Price Forecasting

Traditional approaches include ARIMA (Becerra et al., 2022), GARCH (Garcia and Kristjanpoller, 2019), and VAR/VECM models (Dai, 2025). Machine learning methods—LSTM (Hu et al., 2020; Luo et al., 2022), CNN-LSTM (Li et

al., 2023), XGBoost (Nabavi et al., 2024), and Transformers (Tseng and Nguyen, 2025; Zhao et al., 2025)—have shown improvements over econometric baselines, particularly at longer horizons.

### B. Decomposition-Ensemble Methods

Signal decomposition followed by component-wise forecasting is the most active methodological area. VMD-LSTM (Liu et al., 2020) is, to our knowledge, the most-cited work specifically combining VMD with LSTM forecasting for non-ferrous metals, demonstrating that VMD’s variational formulation avoids the mode mixing issues of EMD/EEMD (Dragomiretskiy and Zosso, 2014). Recent advances include secondary decomposition (Liu et al., 2022; Zhang et al., 2025a), adaptive VMD parameter selection (Zhang and Ke, 2025), and complexity-aware decomposition (Li and Liu, 2026). The large majority of decomposition-ensemble methods treat variables independently, discarding cross-variable information.

### C. Graph Neural Networks for Financial Forecasting

MTGNN (Wu et al., 2020) introduced learned adaptive adjacency matrices for multivariate time series, and our per-band graph construction (Section III-C) follows the same adjacency parameterization applied independently within each frequency band. StemGNN (Cao et al., 2020) operates in the joint spectral-temporal domain. FourierGNN (Yi et al., 2023) unifies spatial and temporal modeling via Fourier operators. In finance, THGNN (Xiang et al., 2023) uses dynamic heterogeneous graphs for stocks, and MDGNN (Qian et al., 2024) introduces multi-relational dynamic graphs.

For commodities specifically, the literature is sparse. Sun et al. (2025) proposed a Multi-view Graph Transformer for copper that constructs a single graph over undecomposed price data—the graph structure does not vary by frequency scale. Hu et al. (2023) applied heterogeneous continual GNN to 49 Chinese futures contracts. Zhai et al. (2024) used a Graph Fourier Transformer for base metals.

### D. VMD Combined with GNN, and Positioning of This Work

A small number of papers combine VMD with GNN, mostly outside finance: traffic forecasting (Ahmad et al., 2024, 2025), air quality (Pei et al., 2022), wind energy (Cheng and Tsai, 2026; Bao et al., 2025), and industrial applications (Yuan, 2025). Within finance, Li and Liu (2026) (“MBTI-Net”) combines VMD decomposition with a GNN for copper futures price forecasting; this is the closest prior work to ours and rules out any claim of being first to combine VMD and GNNs for copper. Critically, MBTI-Net pools the decomposed VMD modes into a single unified graph before applying graph convolution—the graph topology does not vary across frequency bands. Zheng and Zhufu (2025) combine DCT (not VMD) with a cross-scale GNN for single-stock prediction, not multi-commodity relational modeling.

The gap we target is therefore precise: among work that combines signal decomposition with graph-based relational

TABLE I: Positioning against the most directly comparable prior work. “Per-band graph” means a separate learned adjacency per frequency/scale component; “pooled/single” means one graph shared across components or over undecomposed data.

| Work                         | Decomp.? | Graph                              | Domain            |
|------------------------------|----------|------------------------------------|-------------------|
| VMD-LSTM (Liu et al., 2020)  | VMD      | —                                  | metals (uni-var.) |
| MTGNN (Wu et al., 2020)      | —        | learned, single                    | multiv. TS        |
| StemGNN (Cao et al., 2020)   | spectral | learned, single                    | multiv. TS        |
| FourierGNN (Yi et al., 2023) | Fourier  | learned, single                    | multiv. TS        |
| THGNN (Xiang et al., 2023)   | —        | dynamic, single                    | stocks            |
| MDGNN (Qian et al., 2024)    | —        | multi-relational                   | stocks            |
| Sun et al. 2025              | —        | single, undecomp.                  | copper            |
| Hu et al. 2023               | —        | heterog., single                   | futures (49)      |
| Zhai et al. 2024             | Fourier  | learned, single                    | base metals       |
| Zheng & Zhufu 2025           | DCT      | cross-scale, single                | single stock      |
| MBTI-Net (Li and Liu, 2026)  | VMD      | <b>pooled</b>                      | copper            |
| <b>This work</b>             | VMD      | <b>per-band (<math>K=5</math>)</b> | copper (multi.)   |

modeling for commodity or financial time series, we are not aware of any that constructs a *separate* graph per decomposed frequency band across a multi-commodity system, as opposed to (a) pooling bands into one graph (MBTI-Net) or (b) using a single graph over undecomposed data (Sun et al., 2025). Table I makes this positioning checkable against the most directly comparable prior work discussed above, rather than only argued in prose. We do not claim priority over VMD+GNN combinations generally, nor over GNN applications to copper generally.

More importantly, we do not simply assert that the per-band choice is an improvement over MBTI-Net-style pooling and leave it there: we treat it as an empirical question MBTI-Net’s architecture implicitly answers one way (pooling is sufficient) without testing the alternative, and we test both sides of that question directly, against a matched pooled-graph baseline (Section IV-D), at the same data scale. Our result complicates a straightforward reading of MBTI-Net’s design choice: we find no ablation evidence that per-band graphs outperform pooling, and we go further than reporting that negative comparison by diagnosing *why* it comes out this way—the per-band graph-learning mechanism in our model collapsed to near-uniform edge weights during training

(Section V-E1), a specific, quantified failure mode that, to our knowledge, has not previously been documented for this class of learned-adjacency GNN architecture at low node counts. It is not, however, without precedent, and we do not present it as unanticipated: Fountoulakis et al. (2023) prove, and confirm on both synthetic and real graphs, that in a low signal-to-noise (“hard”) regime of the contextual stochastic block model, a graph attention network’s coefficients concentrate on uniform  $1/|\mathcal{N}_i|$  weights, so the attention mechanism reduces to ordinary mean-aggregating graph convolution and cannot distinguish important from unimportant edges. Their setting is attention over a *given* graph in node classification, and their mechanism is the statistical indistinguishability of intra- from inter-class edges rather than an optimization dynamic; ours is a *learned* bilinear adjacency in a forecasting model, degenerating for a different, traceable reason (Section V-E1). The qualitative endpoint — graph attention collapsing to uniform weights when there is too little signal to differentiate edges — is nonetheless the same, which we take as corroborating rather than diminishing the finding. This does not establish that MBTI-Net’s own pooled graph suffers the same failure—we did not have access to retrain or inspect MBTI-Net’s checkpoints—but it is a concrete, mechanistic reason to be skeptical that per-band-versus-pooled is a distinction that reliably matters at this problem scale, and a more informative contribution to that specific architectural question than either an unexamined win or an unexamined loss would be.

### III. METHODOLOGY

#### A. Problem Formulation

Given  $N = 15$  multivariate time series  $\{\mathbf{x}_1, \dots, \mathbf{x}_N\}$  representing copper price and related commodity/macroeconomic variables (Section IV-A), we aim to forecast the copper price return  $r_{t+h} = \ln(p_{t+h}/p_t)$  at horizons  $h \in \{1, 5, 10, 22\}$  trading days, using a lookback window of  $L$  timesteps. We state the exact indexing convention once here since it is easy to get subtly wrong: the window used to predict  $r_{t+h}$  spans timesteps  $t - L, \dots, t - 1$  (strictly before  $t$ , the day the forecast is issued from), not  $t - L + 1, \dots, t$ ; this is the same convention the ARIMA baseline documents explicitly in its implementation, and every model in this study uses it identically, so it does not affect any cross-model comparison, but every forecast horizon is effectively one trading day longer in wall-clock terms than the horizon label alone would suggest. Table II collects the symbols used across this section and Sections V-E1 and V-G, since the notation is dense enough that we found it worth fixing in one place rather than only at each symbol’s first use.

#### B. Variational Mode Decomposition

VMD (Dragomiretskiy and Zosso, 2014) decomposes a signal  $f(t)$  into  $K$  band-limited intrinsic mode functions  $\{u_k\}_{k=1}^K$  by solving:

$$\min_{\{u_k\}, \{\omega_k\}} \sum_{k=1}^K \left\| \partial_t \left[ \left( \delta(t) + \frac{j}{\pi t} \right) * u_k(t) \right] e^{-j\omega_k t} \right\|_2^2 \quad \text{s.t.} \quad \sum_{k=1}^K u_k = f \quad (1)$$

TABLE II: Notation used in Sections III, V-E1, and V-G.

| Symbol                                   | Meaning (fixed value in this paper)                                                                   |
|------------------------------------------|-------------------------------------------------------------------------------------------------------|
| $N$                                      | number of input variables ( $N = 15$ , Section IV-A)                                                  |
| $K$                                      | number of VMD frequency bands ( $K = 5$ ; swept $\{3, 5, 7, 9\}$ in Section V-G)                      |
| $L$                                      | lookback window length ( $L = 60$ trading days)                                                       |
| $T$                                      | total series length (lookback timestep index range)                                                   |
| $d$                                      | graph-embedding dimension ( $d = 16$ , fixed, not HPO-searched)                                       |
| $d_h$                                    | hidden dimension (64 default; 32/64/128 HPO-searched)                                                 |
| $H$                                      | number of GAT attention heads (4 default; 2/4/8 HPO-searched)                                         |
| top- $k$                                 | per-row adjacency sparsification budget ( $k = 7$ of $N = 15$ , i.e. $\lfloor N/2 \rfloor$ )          |
| $\mathcal{H}$                            | forecast horizon set, $\{1, 5, 10, 22\}$ trading days                                                 |
| $\mathbf{A}_k$                           | learned adjacency matrix for band $k$                                                                 |
| $\mathbf{E}_k^{(1)}, \mathbf{E}_k^{(2)}$ | bilinear graph-embedding matrices for band $k$ , $\in \mathbb{R}^{N \times d}$                        |
| $\mathbf{z}^{(k,h)}$                     | band- $k$ representation after GAT+LSTM, for horizon $h$ (own LSTM per band per horizon, Section III) |
| $w_k^h$                                  | attention-fusion weight for band $k$ , horizon $h$ (own fusion module per horizon)                    |
| $\lambda_h$                              | per-horizon loss weight, $\propto 1/\text{Var}[\text{targets}_h]$ , mean-normalized (Section IV-G)    |

where  $\omega_k$  is the center frequency of mode  $k$ . The constrained problem is solved via ADMM in the Fourier domain with penalty parameter  $\alpha$ .

We apply VMD independently to each of the  $N = 15$  input variables with  $K = 5$  modes, yielding a tensor of modes  $\mathbf{U} \in \mathbb{R}^{N \times K \times T}$ . To prevent temporal leakage, decomposition uses a *rolling window, refit at every trading day*: at trading day  $t$ , VMD is recomputed from scratch on the window of the most recent 252 trading days ending at and including  $t$  (signal  $[\max(0, t-251) : t+1]$ ), never using data beyond  $t$ , and only the final-timestep value of each resulting mode is retained as day  $t$ ’s feature. Because refitting happens at every single day rather than periodically, each mode’s assigned series is a genuinely daily-varying signal at the same temporal resolution as the raw price data, not a value held constant across a multi-day block. Because VMD’s ADMM optimization does not guarantee a consistent mode ordering across independent daily fits, each day’s  $K$  modes are re-sorted by ascending converged center frequency ( $\omega_k$ ) before being stored, so mode index 0 is always the lowest-frequency band and mode  $K - 1$  the highest, consistently across the full time series. An earlier version of this pipeline instead used an expanding-window decomposer that refit only every 21 trading days and carried the resulting modes forward (constant) across each 21-day block; that design was found to collapse a 60-day lookback window into as few as three distinct feature values per variable/mode and was replaced by the daily-refit rolling-window design described above before any of the results in this paper were produced. The concern this design addresses is not idiosyncratic to this study: Feng et al. (2026) independently identify temporal leakage in VMD-

based forecasting as a general problem in the electricity-demand literature — noting that decomposing an entire series, or even an entire input subsequence, before forecasting grants the model access to future information — and adopt a sample-wise, rolling-window VMD scheme for the same causality-preserving reason we adopt a daily-refit one here. Decomposed modes and refit metadata (split dates, VMD parameters, the rolling-window length, and a hash of the underlying price data) are cached to disk, with cache invalidation triggered by any change to the underlying data or parameters.

### C. Frequency-Specific Graph Construction

The central architectural choice in VMD-MFGNN is constructing a *separate* graph  $G_k = (V, E_k)$  for each frequency band  $k \in \{1, \dots, K\}$ , where nodes  $V = \{1, \dots, N\}$  represent the  $N = 15$  variables and edges  $E_k$  capture band-specific inter-variable dependencies. This is directly contrasted, via ablation, against a single graph pooled across all  $K$  bands (Section IV-D).

We employ learned adaptive adjacency matrices following Wu et al. (2020), applied independently within each band. For each band  $k$ , two embedding matrices  $\mathbf{E}_k^{(1)}, \mathbf{E}_k^{(2)} \in \mathbb{R}^{N \times d}$  are randomly initialized and jointly optimized:

$$\mathbf{A}_k = \text{softmax} \left( \text{ReLU} \left( \mathbf{E}_k^{(1)} \mathbf{E}_k^{(2)\top} \right) \right) \quad (2)$$

(the implementation applies no  $1/\sqrt{d}$  scaling to the bilinear logits before ReLU/softmax; we state this explicitly because this absence turns out to matter for the collapse mechanism diagnosed in Section V-E1). Each row of  $\mathbf{A}_k$  is then sparsified to its top- $k = 7$  entries (out of  $N = 15$ ), so the realized per-band graph is not fully connected; the complexity analysis in Section III-G accounts for this sparsification. This produces  $K$  adjacency matrices, one per frequency band, each intended to capture dependencies at its own frequency scale. As an ablation-only alternative (Section IV-D), the learned adjacency can be replaced by a fixed, precomputed adjacency built from Pearson correlation between variables' band signals, thresholded at  $|r| > 0.3$  (edges below this magnitude are dropped, following a conventional weak-to-moderate correlation cutoff). We note two implementation details of this fixed graph, since they mean it is not a drop-in magnitude-only replacement for the learned adjacency: the retained entries keep their *signed* correlation value (negative correlations reach the GAT layer as negative edge features, unlike the learned adjacency's post-softmax non-negative weights), and the diagonal is fixed at 1.0 (a self-loop between  $1 \times$  and, given the  $|r| > 0.3$  retention threshold, at most  $\approx 3.3 \times$  larger than any retained off-diagonal entry), whereas the learned path instead receives GAT's own mean-valued self-loops. Both are held constant across all reported correlation-graph results, so they do not confound the correlation-graph ablation's own reported numbers, but a reader comparing it directly against the learned-adjacency mechanism should be aware the two are not scaled identically. The correlation matrices themselves are computed once, from the training split only, and frozen for the remainder of that run (validation and test batches reuse the same frozen matrices

rather than recomputing their own), so this ablation does not leak test- or validation-period correlation structure into the graph it is scored against.

### D. Frequency Band Module

For each band  $k$ , a **FrequencyBandModule** processes the  $k$ -th modes of all  $N$  variables:

**Graph Attention Layer.** At each timestep  $t$ , the mode values  $\mathbf{u}_t^{(k)} \in \mathbb{R}^N$  are projected to hidden dimension  $d_h$  and processed by a multi-head Graph Attention Network. The original formulation of graph attention (Velickovic et al., 2018) is what this architecture is conceptually descended from, but the layer actually trained for every result this paper reports is **GATv2** (Brody et al., 2022), not the original GAT: the implementation uses PyTorch Geometric's `GATv2Conv` (confirmed directly from the trained checkpoints' parameter names, `lin_l/lin_r/a` single att vector, which is `GATv2Conv`'s signature, not the original `GATConv`'s separate `att_src/att_dst`). Brody et al. (2022) show the original GAT's attention is *static* — for a fixed query node, the ranking of attention over its neighbors does not depend on the query, because the query and key projections are applied before, rather than after, the shared nonlinearity — and that swapping the order of the projection and the nonlinearity yields *dynamic*, strictly more expressive attention at the same parameter and computational cost; `GATv2Conv` is that same-cost drop-in:

$$\mathbf{h}_t^{(k)} = \text{GATv2}(\mathbf{u}_t^{(k)}, \mathbf{A}_k) + \mathbf{u}_t^{(k)} \quad (3)$$

with residual connections and layer normalization, stacked over `num_gnn_layers` GAT layers. We write `GATv2`( $\mathbf{u}_t^{(k)}, \mathbf{A}_k$ ) for brevity, but  $\mathbf{A}_k$  does not act as a fixed aggregation weight the way it would in a GCN: the implementation (`GATv2Conv` with `edge_dim=1`) uses  $\mathbf{A}_k$ 's top- $k$ -sparsified entries only to fix which edges exist, passing each kept edge's weight in as a scalar edge feature; the GAT layer then computes its own attention coefficient for every edge from the node features and that edge feature jointly, and it is this learned attention coefficient, not  $\mathbf{A}_k$  directly, that weights the message passed along the edge. Collapse of  $\mathbf{A}_k$  to near-uniform values (Section V-E1) therefore removes the edge-feature signal GAT's attention could otherwise use to differentiate edges, rather than literally forcing uniform aggregation weights, though the two are observed to coincide in practice (Section V-E1). (This paper's earlier-phase,  $N = 8$  robustness checkpoints, Section V-G, predate this `GATv2` swap and use the original, static-attention `GATConv` instead — confirmed the same way, from those checkpoints' `att_src/att_dst` parameter names — so the two GAT variants coexist in this paper across its current- and historical-scope results respectively.)

**Target-node extraction (shared).** The per-timestep, per-variable representations  $\mathbf{h}_1^{(k)}, \dots, \mathbf{h}_L^{(k)} \in \mathbb{R}^{N \times d_h}$  produced by the graph-attention stage above are shared across all four forecast horizons; from them, only the target variable's (copper, node 0) sequence  $\mathbf{h}_{\text{Cu},1}^{(k)}, \dots, \mathbf{h}_{\text{Cu},L}^{(k)} \in \mathbb{R}^{d_h}$  is carried forward into the temporal stage below. This is not an approximation:

the GAT layer above has already mixed every node's information into node 0's representation via message passing, so extracting node 0's sequence here and running the temporal model on it alone is equivalent to running that same LSTM independently on all  $N$  nodes and discarding every output but node 0's, at  $N$ -fold lower cost (Section III-G).

**Temporal Layer (per horizon, not shared). Correction relative to an earlier version of this paper:** the temporal stage is *not* a single LSTM shared across all four forecast horizons. Each forecast horizon  $h \in \mathcal{H}$  instead owns its own independent LSTM per frequency band (confirmed directly from the trained checkpoint behind Table IV, `checkpoints/vmd_mfgnn.pt`, which contains `228 horizon_branches.*` parameters, one `HorizonBranch` submodule per horizon, each with its own  $K$  LSTMs, its own fusion module below, and its own prediction head): only the graph-construction and GAT stages above are shared across horizons; everything downstream is horizon-specific. For band  $k$  and horizon  $h$ :

$$\mathbf{z}_{\text{Cu}}^{(k,h)} = \text{LSTM}_h(\mathbf{h}_{\text{Cu},1}^{(k)}, \dots, \mathbf{h}_{\text{Cu},L}^{(k)}) \in \mathbb{R}^{d_h} \quad (4)$$

so  $K \times |\mathcal{H}| = 5 \times 4 = 20$  independent LSTMs exist in the model at this paper's default settings (rather than  $K = 5$  shared ones, as an earlier version of this section described), each the same size as a single shared LSTM would have been. The rationale for this design (rather than a cost argument) is that a single shared temporal representation previously had to serve both the 1-day, noise-dominated horizon and the 22-day, trend-dominated horizon simultaneously; giving each horizon its own LSTM lets it specialize, while the graph/GAT stage stays shared precisely so its already-weak gradient signal (Section V-E1) is not diluted further by a four-way split.

Algorithm 1 summarizes the full forward pass just described, from raw multivariate input through the shared per-band graph construction and GAT processing to the per-horizon temporal, fusion, and prediction steps detailed in the remainder of this section, and Figure 1 renders the same forward pass as a block diagram with tensor shapes, the shared-versus-per-horizon split, and the per-band repetition made explicit. The two are intended to be read together: every block in the figure corresponds to one or more numbered lines of the algorithm, and the figure adds no component the algorithm does not specify. It describes the architecture as designed; whether the per-band graph mechanism (lines 4–7) learns the differentiated structure it is intended to is an empirical question taken up separately in Section V-E1.

### E. Attention-Based Mode Fusion

The  $K$  band-level representations are fused via a learned attention mechanism — **one instance of this mechanism per horizon**  $h \in \mathcal{H}$ , not one shared instance:

$$w_k^h = \frac{\exp(\mathbf{q}_h^\top \mathbf{W}_h \mathbf{z}_{\text{Cu}}^{(k,h)})}{\sum_{j=1}^K \exp(\mathbf{q}_h^\top \mathbf{W}_h \mathbf{z}_{\text{Cu}}^{(j,h)})}, \quad \mathbf{z}_{\text{fused}}^h = \sum_{k=1}^K w_k^h \cdot \mathbf{z}_{\text{Cu}}^{(k,h)} \quad (5)$$

where  $\mathbf{q}_h$  is horizon  $h$ 's own learnable query vector and  $\mathbf{W}_h$  is horizon  $h$ 's own single learned linear projection (not a multi-layer network) applied to each band representation before

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### Algorithm 1 VMD-MFGNN forward pass

---

**Require:** Raw multivariate window  $\{\mathbf{x}_1, \dots, \mathbf{x}_N\}$  for  $N = 15$  variables over lookback  $L = 60$ ; number of frequency bands  $K = 5$ ; adjacency sparsification parameter top- $k = 7$ ; embedding dimension  $d$ ; hidden dimension  $d_h$ ; number of GAT heads  $H$ ; forecast horizons  $\mathcal{H} = \{1, 5, 10, 22\}$

**Ensure:** Predicted returns  $\{\hat{r}_{t+h}\}_{h \in \mathcal{H}}$

- 1: **VMD decomposition:** for each variable  $n = 1, \dots, N$ , decompose the rolling window ending at  $t - 1$  (i.e. the  $L$  most recent timesteps strictly before the day the forecast is issued from, following this paper's dataset indexing convention) into  $K$  band-limited modes via leakage-safe daily-refit VMD, sort modes by ascending center frequency  $\omega_k$ , and retain the final-timestep value to form  $\mathbf{U} \in \mathbb{R}^{N \times K \times L}$
  - Shared stage (identical for every horizon):**
  - 2: **for**  $k = 1$  to  $K$  **do**
  - 3:   **Graph construction:** initialize (or retrieve) embeddings  $\mathbf{E}_k^{(1)}, \mathbf{E}_k^{(2)} \in \mathbb{R}^{N \times d}$
  - 4:   Compute adjacency logits  $\mathbf{A}_k \leftarrow \text{softmax}(\text{ReLU}(\mathbf{E}_k^{(1)} \mathbf{E}_k^{(2)\top}))$
  - 5:   Sparsify: retain only each row's top- $k = 7$  entries of  $\mathbf{A}_k$ , zeroing the rest
  - 6:   **Band processing:** project band- $k$  mode values  $\mathbf{u}_t^{(k)} \in \mathbb{R}^N$  to hidden dimension  $d_h$
  - 7:   **for**  $t = 1$  to  $L$  **do**
  - 8:      $\mathbf{h}_t^{(k)} \leftarrow \text{GATv2}(\mathbf{u}_t^{(k)}, \mathbf{A}_k) + \mathbf{u}_t^{(k)}$  (residual, layer norm, stacked over `num_gnn_layers` GAT layers)
  - 9:   **end for**
  - 10:   Extract copper's per-timestep sequence  $\mathbf{h}_{\text{Cu},1}^{(k)}, \dots, \mathbf{h}_{\text{Cu},L}^{(k)}$  (node 0 of  $\mathbf{h}_1^{(k)}, \dots, \mathbf{h}_L^{(k)}$ )
  - 11: **end for**
  - Per-horizon stage (own LSTMs, fusion, and head for each  $h$ ):**
  - 12: **for** each horizon  $h \in \mathcal{H}$  **do**
  - 13:   **for**  $k = 1$  to  $K$  **do**
  - 14:      $\mathbf{z}_{\text{Cu}}^{(k,h)} \leftarrow \text{LSTM}_h(\mathbf{h}_{\text{Cu},1}^{(k)}, \dots, \mathbf{h}_{\text{Cu},L}^{(k)})$  (own LSTM per band, per horizon)
  - 15:   **end for**
  - 16:   **Mode fusion:** for  $k = 1, \dots, K$ , compute  $w_k^h \leftarrow \exp(\mathbf{q}_h^\top \mathbf{W}_h \mathbf{z}_{\text{Cu}}^{(k,h)}) / \sum_{j=1}^K \exp(\mathbf{q}_h^\top \mathbf{W}_h \mathbf{z}_{\text{Cu}}^{(j,h)})$
  - 17:    $\mathbf{z}_{\text{fused}}^h \leftarrow \sum_{k=1}^K w_k^h \cdot \mathbf{z}_{\text{Cu}}^{(k,h)}$
  - 18:    $\hat{r}_{t+h} \leftarrow \text{MLP}_h(\mathbf{z}_{\text{fused}}^h)$
  - 19: **end for**
  - 20: **return**  $\{\hat{r}_{t+h}\}_{h \in \mathcal{H}}$
- 

the query dot product;  $\mathcal{H}$  horizons therefore instantiate  $\mathcal{H}$  independent copies of this fusion mechanism, each seeing only its own horizon's band representations. The attention weights  $\{w_k^h\}$  are intended to be interpretable as revealing which frequency bands contribute most to horizon  $h$ 's prediction; Section V reports how peaked or uniform they turn out to be in practice.



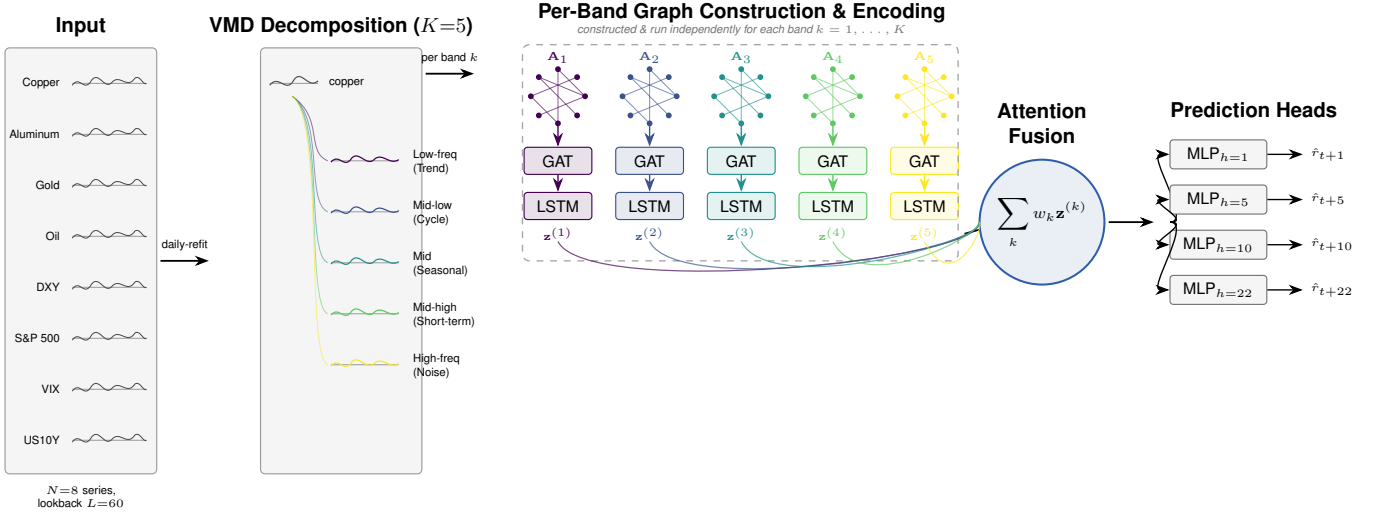


Fig. 1: VMD-MFGNN forward pass. **Note: this figure’s underlying artwork was produced before the per-horizon-branch correction described in the surrounding text and has not yet been regenerated; the caption below describes the architecture correctly, but the rendered diagram itself may still depict a single shared LSTM/fusion stage and should be read alongside Algorithm 1, not in place of it, until it is regenerated.** An  $N = 15$  variable window of length  $L = 60$  trading days (copper and 14 related commodity/macro series, Section IV-A; the icon shows the general per-band mechanism, not one panel per variable) is decomposed by leakage-safe daily-refit VMD into  $K = 5$  frequency-band modes, reusing the same band names and colors as Figure 7. **The five colored node-link icons are this paper’s central architectural claim made visible:** a separate learned graph  $\mathbf{A}_k$  is constructed independently for each band (dashed group), each with its own bilinear embedding pair  $\mathbf{E}_k^{(1)}, \mathbf{E}_k^{(2)}$  (Eq. 2), row-sparsified to top- $k = 7$  of  $N = 15$  edges, and fed to that band’s own GAT (Eq. 3). This per-band graph+GAT stage is the only part of the model shared across forecast horizons: downstream of it, each of the four horizons has its own independent LSTM per band (Eq. 4), its own learned-query attention fusion across the five resulting band representations, and its own prediction head. Algorithm 1 gives the same forward pass line-by-line; the pooled-graph ablation of Section IV-D replaces the five per-band graphs shown here with a single graph shared across bands (Figure 3), but — unlike `full_model` — still uses that earlier, single-shared-LSTM-across-horizons design rather than the per-horizon-branch design described here (Section IV-D discloses this confound explicitly). Section V-E1 shows that, once trained, all five  $\mathbf{A}_k$  collapse to a near-uniform graph indistinguishable from an untrained one.

## F. Prediction and Training

Separate two-layer MLP heads (one per horizon, `Linear`→`ReLU`→`Dropout`→`Linear`, not a single linear layer) map each horizon’s own fused representation  $\mathbf{z}_{\text{fused}}^h$  to that horizon’s prediction:  $\hat{r}_{t+h} = \text{MLP}_h(\mathbf{z}_{\text{fused}}^h)$ . The model is trained end-to-end with a per-horizon-weighted MSE loss summed across the four horizons (the per-horizon weights  $\lambda_h$  are defined and motivated in Section IV-G; an earlier version of this equation omitted them and implied an unweighted sum, which does not match the trainer):

$$\mathcal{L} = \sum_{h \in \mathcal{H}} \lambda_h \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} (r_{t+h} - \hat{r}_{t+h})^2 \quad (6)$$

The best-validation-loss checkpoint is selected and used for all reported models (Section IV-G).

## G. Complexity Analysis

Let  $N = 15$  be the number of variables (Section IV-A),  $T = L$  the lookback window length ( $L = 60$  trading days throughout this paper, Section IV-G),  $K = 5$  the number of VMD modes,  $|\mathcal{H}| = 4$  the number of forecast horizons,  $d_h$  the hidden dimension, and  $H$  the number of GAT attention heads.

Within one frequency band, the learned adjacency  $\mathbf{A}_k$  is top- $k$  sparsified to each node’s  $k = 7$  highest-weight outgoing edges (Section III-C) before being passed to the GAT layer, so the graph each band’s GAT operates on has  $\mathcal{O}(Nk)$  edges rather than  $\mathcal{O}(N^2)$ ; we report the sparsified-graph cost here rather than the looser fully-connected upper bound. This gives per-timestep GAT cost  $\mathcal{O}(Nk d_h / H)$  for the attention computation and  $\mathcal{O}(N d_h^2)$  for the linear projections, repeated over  $T$  timesteps and `num_gnn_layers` GAT layers; this GAT stage is shared across all  $|\mathcal{H}|$  forecast horizons (Section III), so its cost is incurred once, not once per horizon. **Correction relative to an earlier version of this section:** the temporal stage does not cost  $\mathcal{O}(NT d_h^2)$  applied “independently per variable.” The LSTM only ever runs on the target (copper) node’s already-GAT-mixed sequence (Section III), and it does so once per band *and* once per forecast horizon, since each horizon owns its own independent per-band LSTM rather than sharing one: its real cost is  $\mathcal{O}(|\mathcal{H}| K T d_h^2)$ , a single node’s sequence length  $T$  processed  $K \times |\mathcal{H}| = 20$  times (at this paper’s default  $K = 5$ ,  $|\mathcal{H}| = 4$ ) rather than  $N = 15$  times. Combining the shared graph/GAT cost (incurred once, across all  $K$  bands) with the per-horizon temporal cost, VMD-MFGNN’s total

forward-pass cost is  $\mathcal{O}(K(NkTd_h + NTd_h^2) + |\mathcal{H}|KTd_h^2)$ : the first term is  $K \times$  the cost of a single-graph architecture with otherwise identical per-band capacity (the direct computational price of per-band graph construction, and the reason the pooled-graph ablation of Section IV-D is evaluated as a single-graph, single-pass alternative rather than dismissed a priori as architecturally uninteresting), and the second term is the added cost of giving each forecast horizon its own temporal pathway rather than sharing one, which is small in practice since it acts on a single node’s sequence rather than all  $N$ .

Two distinct trained instances of the VMD-MFGNN architecture appear in this paper, and their parameter counts must not be conflated. The main comparison (Section V) uses the HPO-tuned configuration selected by the search in Section IV-C ( $d_h = 64$ ,  $H = 2$ , `num_gnn_layers`=3), which has **1,491,748** trainable parameters, counted directly from the saved, trained model (`checkpoints/vmd_mfgnn.pt`). The ablation study (Section IV-D) instead uses the untuned default configuration ( $d_h = 64$ ,  $H = 4$ , `num_gnn_layers`=2, per Section IV-G) for its `full_model` row, which has **1,448,548** trainable parameters, likewise counted directly from its own saved, trained checkpoint (`checkpoints/full_model.pt`). HPO is applied only to the main-table model, never to the ablation variants (Section IV-C); consequently the ablation table’s `full_model` is a genuinely different model instance than the main table’s VMD-MFGNN row, not the same trained model reused across tables, though at this paper’s real  $N = 15$  data scale the two happen to land within 3% of each other in raw parameter count, unlike the far larger gap this comparison showed at the smaller variable count an earlier phase of this project used. This is a deliberate, disclosed design choice, not an inconsistency: re-running HPO separately for every ablation variant was judged infeasible within the project’s compute budget, and holding the ablation variants at one fixed, shared default configuration keeps the ablation comparison internally consistent even though it is not directly comparable in absolute terms to the main table.

Within the ablation study, capacity is controlled two ways for the no-VMD and pooled-graph variants, and the resulting real parameter counts (from the trained checkpoints) are reported here rather than estimated: a *matched-dimension* variant that keeps the full model’s  $d_h$  fixed and lets parameter count fall out of the architecture (`pooled_graph_matched_dim`: 93,034 params,  $-93.6\%$  vs. `full_model`; `no_vmd_raw_price_matched_dim`: 309,988 params,  $-78.6\%$ ), and a *matched-parameter* variant that increases  $d_h$  until total parameter count is approximately restored to `full_model`’s (`pooled_graph_matched_params`: 1,452,010 params,  $+0.24\%$ ; `no_vmd_raw_price_matched_params`: 1,464,884 params,  $+1.13\%$ ). The *matched-parameter* variants land within about 1% of `full_model`’s parameter count, confirming the capacity-matching procedure worked as intended at real data scale. `correlation_graph` (1,446,148 params,  $-0.17\%$ ), which keeps the full per-band architecture and only replaces the learned adjacency with a fixed correlation-threshold graph, is essentially

parameter-matched to `full_model` by construction (it omits only the small per-band adjacency-embedding matrices). The two graph-fix variants (Section IV-D, item (e)) are parameter-matched to `full_model` almost exactly (`full_model_graphfix`: 1,448,548 params, identical; `full_model_graphfix_temp`: 1,448,553 params,  $+5$  for its five per-band log-temperature scalars), since the fix changes how the graph embeddings are normalized and regularized, not their dimensionality.

## IV. EXPERIMENTAL SETUP

### A. Data

We use daily data from January 2010 to December 2025. The target variable is COMEX copper futures (HG=F). This paper’s real, current input scope is  $N = 15$  variables, sourced from two providers: 11 from Yahoo Finance and 4 from FRED (two daily, two monthly, the latter point-in-time joined with a publication lag to avoid leaking a not-yet-published macro print into an earlier trading day). This supersedes an earlier, smaller  $N = 8$  Yahoo-Finance-only scope used at an earlier phase of this project (retained in Section V-D, Section V-F’s dedicated graph-fix experiment, and Section V-G’s 19-cell robustness suite, all explicitly labeled as earlier-phase there); every other number in this paper describes the real  $N = 15$  scope defined here. The 11 Yahoo Finance series are: copper (HG=F, target), gold (GC=F), crude oil / WTI (CL=F), US Dollar Index (DX-Y.NYB), S&P 500 (^GSPC), VIX (^VIX), the US 10-year Treasury yield (^TNX), silver (SI=F), the Chilean peso (CLP=X, a producer-currency signal, Chile being the world’s largest copper producer), Freeport-McMoRan equity (FCX, a copper-supply-side signal), and the iShares China Large-Cap ETF (FXI, a China-demand proxy). The 4 FRED series are: Moody’s Baa corporate bond yield spread (BAA10Y) and the 10-year TIPS real yield (DFI10), both daily; and the Producer Price Index: All Commodities (PPIACO) and Industrial Production Index (INDPRO), both monthly, forward-filled after their point-in-time publication date. Aluminum (ALI=F) and zinc/nickel proxies were evaluated and dropped; the reasons are disclosed in Section VI (items 1 and 7) rather than repeated here.

**Trading-calendar alignment.** COMEX copper futures, the S&P 500, CBOE VIX, and the US 10-year yield do not share an identical exchange holiday calendar. The 11 daily (Yahoo Finance and FRED-daily) series are joined on the union of their trading days, forward-filled for gaps of up to 5 trading days (`ffill(limit=5)`), covering the isolated single-exchange holidays and short data-vendor gaps this scope exhibits), and any row still containing a missing value after that limited fill is dropped rather than filled further; this 5-day limit applies only to the daily series. The two monthly FRED series (PPIACO, INDPRO) are instead point-in-time joined with a publication-lag offset and then forward-filled across each month, never reindexed-and-filled directly, so no not-yet-published monthly print is ever visible to an earlier trading day. This keeps every reported day’s feature vector either a genuine same-day quote or a bounded, publication-safe carry-forward, never an unbounded or leaking stale value, and applies identically to training, validation, and test data.



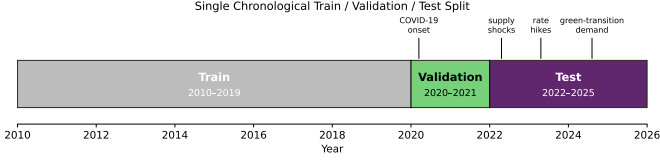


Fig. 2: The single chronological train/validation/test split used throughout this paper, with the named regime events inside each window (Section VI). The split is not walked forward or resampled; this figure makes exactly what one split covers inspectable at a glance.

**Split.** A single chronological train/validation/test split is used: training (2010–2019), validation (2020–2021, includes COVID), test (2022–2025, includes supply shocks, rate hikes, and green-transition demand effects). This is *not* a walk-forward or rolling-origin evaluation with multiple folds; the implications for statistical testing are discussed in Section VI. Figure 2 shows this split on a timeline.

### B. Baselines

We compare VMD-MFGNN against seven baselines: ARIMA, XGBoost, LSTM, Transformer, VMD-LSTM, SimpleMTGNN, and MTGNN, spanning econometric, machine-learning, deep-learning, decomposition-ensemble, and graph-based categories. All seven run with standard, reasonable, hand-set configurations; none receive hyperparameter optimization (Section IV-C). ARIMA selects its  $(p, d, q)$  order by AIC over a small grid per series; XGBoost uses 500 estimators, max depth 6, learning rate 0.05; LSTM, Transformer, VMD-LSTM, and SimpleMTGNN share a common training loop (hidden dimension 64, 2 recurrent/GNN layers where applicable, Adam at learning rate  $10^{-3}$ , up to 100 epochs with patience 20, lookahead 60) and differ only in their architecture-specific layers, described next. MTGNN is trained under the same shared protocol (Adam, learning rate  $10^{-3}$ , weight decay  $10^{-5}$ , gradient clipping at 1.0, the same early-stopping rule as every other baseline) but keeps its own architecture’s official default hyperparameters (`gc_n_depth= 2`, `node_dim= 40`, `dropout= 0.3`, `conv_channels=residual_channels= 32`, `skip_channels= 64`, `end_channels= 128`, 3 layers, `propalpha= 0.05`, `tanhalph= 3`) rather than this project’s shared hidden-dimension convention, since deviating from them would compromise the reproduction claim below.

We are explicit that “SimpleMTGNN” and “VMD-LSTM” are our own simplified reimplementations of the design ideas in Wu et al. (2020) and the VMD-LSTM literature respectively, **not the original authors’ code**, since neither published implementation is a plug-and-play fit for this study’s variable set and multi-horizon output. SimpleMTGNN keeps MTGNN’s core idea — a learned bilinear node-embedding adjacency feeding graph-based spatial mixing — but omits the original’s saturated top-k graph-learning layer, mix-hop propagation, dilated-inception temporal convolution, and skip connections, using instead a single linear spatial mix and two plain dilated causal convolutions; it should be

read as a graph-learning baseline in the same architectural family as MTGNN, not a reproduction of its reported results. **MTGNN itself, by contrast, is a genuine architectural reproduction of Wu et al. (2020)**, ported from the authors’ official implementation with all four of its components intact (saturated top-k graph learning, mix-hop propagation, dilated-inception temporal convolution, and output skip connections); it is included alongside SimpleMTGNN specifically so that the simplified baseline’s results are not the only graph-learning comparison point in this study. VMD-LSTM decomposes the input with the same VMD front end as VMD-MFGNN but feeds all input variables jointly into a single per-mode LSTM (rather than one univariate LSTM per decomposed variable, the more common convention in the VMD-LSTM literature), so that it isolates the graph-construction step specifically as VMD-MFGNN’s point of difference from a non-graph VMD+LSTM alternative, at the cost of not being a strict reproduction of any single prior VMD-LSTM paper’s exact architecture. We flag here that two baselines’ cells are visibly pathological rather than merely weaker (LSTM at  $h = 22$ : RMSE 0.3607 against  $\approx 0.08$ –0.10 for the best-behaved models,  $R^2 = -22.70$ ; Transformer at  $h = 10$ : RMSE 0.1394,  $R^2 = -5.64$ , Table V), and Section V-B revisits what this means for the significance comparisons that involve them.

### C. Hyperparameter Optimization

VMD-MFGNN, the proposed model, is tuned via a lightweight Optuna study (TPE sampler with a median pruner), searching hidden dimension  $\in \{32, 64, 128\}$ , number of attention heads  $\in \{2, 4, 8\}$  (constrained so hidden dimension is divisible by the head count), learning rate (log-uniform,  $10^{-4}$  to  $10^{-2}$ ), dropout (uniform, 0.05 to 0.3), and number of GNN layers  $\in \{1, 2, 3\}$ , configured for up to 15 trials of 25 epochs each. Table III tabulates this search space alongside the selected best trial. In the run reported in this paper, the study produced 11 rows against the configured 15 trials: 8 COMPLETE, 2 PRUNED by the median pruner before completing their allotted epochs, and 1 left in RUNNING state with no recorded value, never resolved (most likely a Colab disconnect mid-trial) before the search budget was exhausted. We report all three states rather than folding the orphaned trial into either of the other two: selection among named, completed configurations is therefore effectively over 10 usable trials (8 complete plus 2 pruned), not 11 and not the originally configured 15. The best trial (validation MSE = 0.001691) selected **hidden dimension = 64**, **attention heads = 2**, **learning rate  $\approx 0.00101$** , **dropout  $\approx 0.199$** , **GNN layers = 3**. We report this search plainly as weak and close to noise rather than as strong evidence of a clearly optimal configuration: the best trial sits about 12% below the next-best completed trial (validation MSE 0.001929), but the remaining 7 completed trials cluster tightly within about 5% of each other (0.001929–0.002032) with no visible trend against any single searched hyperparameter, which is not the signature of a well-resolved optimum. This configuration is used for VMD-MFGNN throughout the main results (Table IV) and significance testing (Table VI); it is *not* used in the ablation

TABLE III: Main Optuna search space (Section IV-C) and the selected best trial (validation MSE = 0.001691, 11 of 15 configured trials recorded: 8 COMPLETE, 2 PRUNED, 1 orphaned RUNNING trial with no usable value).

| Hyperparameter      | Search space                   | Best trial        |
|---------------------|--------------------------------|-------------------|
| hidden dim. $d_h$   | {32, 64, 128}                  | 64                |
| attention heads $H$ | {2, 4, 8}                      | 2                 |
| learning rate       | log-U( $10^{-4}$ , $10^{-2}$ ) | $\approx 0.00101$ |
| dropout             | U(0.05, 0.3)                   | $\approx 0.199$   |
| GNN layers          | {1, 2, 3}                      | 3                 |

study (Section IV-D), which fixes all variants, including its own `full_model` row, at the untuned default configuration (Section IV-G) instead. The seven baselines are *not* tuned; they run at standard/hardcoded settings. This asymmetry is a deliberate, disclosed methodological choice rather than an oversight: tuning all eight models under a full hyperparameter search would multiply compute cost roughly by the trial count and was judged infeasible within this project’s compute and time budget, and tuning only the proposed model against standard baselines is common, defensible practice in comparison papers under such constraints, provided it is disclosed (Section VI). HPO results (best parameters and the full trial history, including pruned and orphaned trials) are saved for reproducibility.

#### D. Ablation Studies

Eight variants isolate the contribution of each architectural component, all trained at the untuned default configuration (Section IV-G, Section III-G)—HPO (Section IV-C) is applied only to the main-table VMD-MFGNN, never to any ablation variant. The ablation results below are reported as point estimates per horizon; a Diebold-Mariano test on this paper’s single central ablation comparison (`full_model` vs. `pooled_graph_matched_params`) is reported alongside Table VIII in Section V-C, and the remaining ablation pairs are left untested, for the reasons given in Section IV. A separate five-seed repetition of the two variants that carry RQ1, run later at an earlier project phase’s own  $N = 8$  tuned configuration (Section V-G’s staleness disclosure) and therefore not directly comparable to this table, supplies a seed-level paired test and is reported in Section V-G.

Figure 3 shows the architectural difference RQ1 asks about: variant (a), the full model of Figure 1, learns  $K = 5$  independent adjacency matrices and runs  $K$  separate GAT+LSTM branches before fusing them; variant (c) pools the  $K$  raw per-band VMD values for each variable through a learned linear projection (`nn.Linear(K, 1)`) before any GAT/LSTM encoding, then learns a single shared adjacency and runs one GAT+LSTM branch over that pooled scalar signal. We are explicit that this pools raw band *values*, not learned band *representations*, since the latter would imply pooling happens after per-band encoding rather than instead of it. The remaining five variants (the two capacity settings of b, d, and the two capacity settings of c) are described below.

- **(a) Full model:** VMD decomposition into  $K = 5$  bands, with a separate learned graph per band, at the untuned

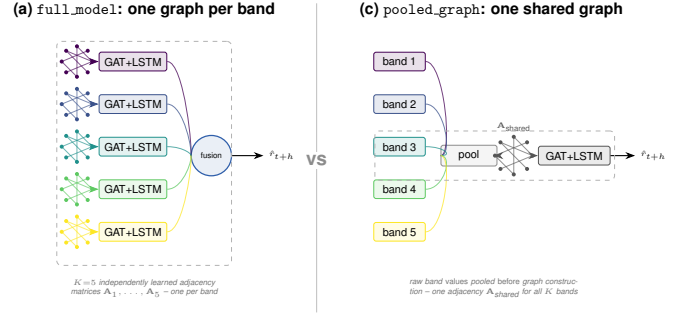


Fig. 3: The RQ1 comparison made visible: (a) `full_model` builds a separate learned graph per frequency band; (c) `pooled_graph` pools raw band *values* (not learned representations) via a linear projection before constructing a single shared graph. Section V-E1 shows that in the trained model, (a)’s five graphs collapse to a near-uniform adjacency indistinguishable from one another, which is why the ablation in Table VIII does not show a clean advantage for (a) over (c).

default configuration (1,448,548 parameters). This is a distinct trained instance from the HPO-tuned VMD-MFGNN reported in Table IV (1,491,748 parameters); see Section III-G.

- **(b) No-VMD, raw price:** A genuine raw-price GAT+LSTM model with a single learned graph, using a dedicated raw-price dataloader that never touches VMD-decomposed input, reported at two capacity settings: *matched-dim* (same hidden dimension as the full model, 309,988 params,  $-78.6\%$ ) and *matched-params* (hidden dimension increased to restore parameter count, 1,464,884 params,  $+1.13\%$ ). This isolates the contribution of decomposition itself, independent of the per-band-graph question, at two different capacity assumptions.
- **(c) Pooled graph:** VMD decomposition retained, but the  $K$  raw band *values* are pooled (via a learned linear projection, before any GAT/LSTM encoding) into a single unified signal before a single graph is built over it, rather than building  $K$  separate graphs. This is the ablation that most directly supports (or fails to support) the paper’s central architectural claim against MBTI-Net-style pooling. Reported at two capacity settings: *matched-dim* (93,034 params,  $-93.6\%$ ) and *matched-params* (1,452,010 params,  $+0.24\%$ ). The *matched-params* setting lands within about 1% of the full model’s parameter count, so the (a)-vs-(c, *matched-params*) comparison is the cleanest available test of graph structure with capacity approximately controlled; the *matched-dim* setting is retained alongside it because it reflects the capacity a pooled architecture has *by default* at equal hidden dimension, which is itself an architecturally meaningful comparison.
- **(d) Correlation graph:** The learned per-band adjacency is replaced with a fixed adjacency built from Pearson correlation between variables’ band signals, thresholded at  $|r| > 0.3$  (essentially parameter-matched to the

full model). This isolates the contribution of end-to-end learned adjacency versus a fixed, precomputed graph, holding the per-band architecture fixed.

- **(e) Graph-fix variants:** Two further variants apply the anti-collapse fix of Section V-F (unit-normalized graph embeddings, excluded from weight decay) directly inside the ablation study rather than only in the separate graph-fix experiment: `full_model_graphfix` applies the fix alone, and `full_model_graphfix_temp` applies the same fix plus a learnable, per-band softmax temperature ( $\tau = \exp(\log\_temperature)$ , itself excluded from weight decay by the same mechanism). Both are included here specifically to ask whether the fix that mechanically resolves the embedding collapse also helps forecasting accuracy, which the graph-fix experiment’s own accuracy numbers do not by themselves settle. **Naming note:** `full_model_graphfix` here is a distinct, newer checkpoint from this variant of the same name in the dedicated graph-fix experiment of Section V-F — this ablation’s copy is trained at this paper’s real data scale and stops at epoch 29 (Copper\_Paper1.2/checkpoints/full\_model\_graphfix\_epoch29) while Section V-F’s own `full_model_graphfix` is an earlier-phase run trained to epoch 57 under a different `min_epochs` re-run history; the two share an identifier because both preserve the exact checkpoint filename produced by the fix, not because they are the same trained model.

**A further architectural confound in the RQ1 comparison, disclosed rather than hidden.** `full_model` (and every other VMD-based per-band or no-VMD variant above) instantiates VMD-MFGNN’s current, per-horizon-branch design (Section III): a graph+GAT stage shared across all four horizons, followed by an independent LSTM, fusion module, and prediction head per horizon. `pooled_graph_matched_dim` and `pooled_graph_matched_params` (item (c) above), by contrast, are built on `PooledGraphMFGNN`, a separate model class that still uses this paper’s earlier, single-shared-LSTM-across-all-horizons design. We confirmed this directly from the two variants’ saved checkpoints rather than inferring it from the code alone: `full_model`’s checkpoint contains 228 `horizon_branches.* parameters` (one `HorizonBranch` submodule per horizon, each with its own  $K$  LSTMs), while `pooled_graph_matched_params`’s checkpoint contains a single, shared `band_module.lstm`. This means the paper’s central RQ1 comparison (`full_model` vs. `pooled_graph_matched_params`, Section V-C) is confounded along two architectural axes at once — per-band versus pooled graph construction, and per-horizon versus shared temporal modeling — rather than the single clean axis (graph structure alone) it is framed as isolating. We disclose this rather than re-run the ablation under a temporal-design-matched `PooledGraphMFGNN`, which was outside the scope of this correctness pass; the reported numbers themselves are unchanged and real, but the comparison should be read as bearing on “per-band graph

construction bundled with per-horizon temporal modeling” versus “pooled graph construction bundled with shared temporal modeling,” not on graph structure in isolation. The same confound applies to `pooled_graph_matched_dim` and to the five-seed repetition of these two variants reported in Section V-G, but not to the no-VMD variants (item (b) above), which we separately confirmed are themselves built on the current per-horizon-branch `VMDMFGNN` class (`num_modes=1`), not `PooledGraphMFGNN`.

### E. Statistical Testing

For the baseline comparison (VMD-MFGNN versus each of the seven baselines, Section IV-B), we apply the Diebold-Mariano test (Diebold and Mariano, 1995) to each pair of forecast error series, per horizon, testing whether VMD-MFGNN’s forecast errors are significantly different from each baseline’s. This test is applicable here because the same test-set forecasts, produced under the same single split, provide a valid paired sample of loss differentials over time. The test statistic is computed on the loss differential  $d = e_{\text{VMD-MFGNN}}^2 - e_{\text{baseline}}^2$ , divided by the square root of a heteroskedasticity-and-autocorrelation-consistent (HAC) variance estimate that sums  $d$ ’s sample autocovariances at lags  $1, \dots, \text{horizon} - 1$  with equal (rectangular-kernel) weight; this is a simpler, unweighted HAC-style correction, not the Bartlett-weighted kernel of the classical Newey-West estimator, and we describe it as such here to match the implementation exactly rather than name it Newey-West. Consequently a *negative* DM statistic indicates VMD-MFGNN’s squared error was lower (VMD-MFGNN better) and a *positive* DM statistic indicates the baseline’s squared error was lower (baseline better), and this sign convention is used without repeating it per cell in Table VI. Because 28 DM tests are run (seven baselines  $\times$  four horizons), we additionally apply the Holm-Bonferroni step-down procedure (Holm, 1979) to control the family-wise error rate across all 28 tests at  $\alpha = 0.05$ , and report both the raw and Holm-corrected significance in Table VI (Section V-B). An earlier version of this paper stated the DM test could not be applied to the ablation comparisons at all, reasoning from the absence of walk-forward folds or repeated seeds; that reasoning was itself mistaken, since the DM test’s resampling unit is time within a single split (the same paired daily forecast-error series used for the baseline comparison above), not folds or seeds, and a single chronological split is exactly what the baseline comparison above also uses. The real obstacle was practical, not statistical: per-sample predictions were not archived for the ablation study’s variants until a later fix (Section IV-G) added this. With that fix, we now apply the DM test to this paper’s single most important ablation comparison — `full_model` versus `pooled_graph_matched_params`, the capacity-matched pairing that most directly targets RQ1 — and report it in Section V-C alongside Table VIII. We did not extend it to the remaining ablation pairs (no-VMD and correlation-graph variants, or the matched-dim capacity setting), since RQ1 is the comparison this paper’s central claim depends on and we did not want to run a large additional battery of tests without a correction plan to match; this remains a

direction for future work rather than a claimed impossibility. Where seeds *are* available — the five-seed repetition of the two RQ1 variants at an earlier project phase’s own  $N = 8$  tuned configuration (historical, Section V-G) — we separately report paired  $t$ - and Wilcoxon signed-rank tests over seeds, which is complementary evidence at a different, earlier-phase configuration, not a substitute for the DM test above.

#### F. Evaluation Metrics

RMSE, MAE, MASE, SMAPE, and Directional Accuracy (DA) are reported for the main comparison and the ablations, per horizon. SMAPE (Symmetric Mean Absolute Percentage Error,  $200 \times \text{mean}(|y - \hat{y}| / (|y| + |\hat{y}|))$ , bounded to  $[0, 200]\%$ ) is reported instead of the more familiar MAPE: our evaluation code also computes MAPE, but on this near-zero return series MAPE is numerically unstable — its denominator is  $|y|$  alone, so predictions near a true return of zero drive individual terms, and the mean, toward extreme or infinite values, which is why our MAPE computation already has to mask out true values with  $|y| \leq 10^{-8}$  before averaging. SMAPE’s symmetric denominator ( $|y| + |\hat{y}|$ ) does not have this failure mode and stays bounded regardless of how close the true or predicted return is to zero, so it is the percentage-scale metric reported throughout this paper’s tables. MASE (Mean Absolute Scaled Error, Hyndman and Koehler, 2006) is added because it is the standard scale-free accuracy metric in the forecasting-competition literature (M4, M5) and, unlike SMAPE and MAPE, compares each model directly against an explicit naive benchmark rather than only rescaling the error by the target’s own magnitude, so its absence would be conspicuous to a forecasting-literature reviewer. It is computed as  $\text{MASE} = \text{mean}(|y - \hat{y}|) / Q_h$ , where  $Q_h$  is the mean absolute error of a one-step-ahead naive/persistence forecast on the TRAINING split only (never test or validation data, since this is a fixed scaling constant, not a test-set quantity); values below 1.0 beat that in-sample naive benchmark. Because this paper’s targets are  $h$ -day-ahead forward log returns sampled at every trading day, consecutive daily entries of a given horizon’s target series overlap by  $h - 1$  days, so the naive-persistence forecast is not simply the previous daily entry:  $Q_h$  compares each  $h$ -day return to the immediately preceding, *non-overlapping*  $h$ -day return ( $h$  trading days earlier), which is the correct one-step-ahead-in-the-target-series analogue at each horizon and grows the denominator with horizon roughly as  $\sqrt{h}$ , as it should for returns compounded over longer windows. This is distinct from this paper’s zero-forecast baseline row (predicting a return of exactly zero, Section V-A) — MASE’s naive-persistence benchmark predicts “no change from the previous period,” not zero. DA is computed as  $\text{mean}(\text{sign}(r_{t+h}) = \text{sign}(\hat{r}_{t+h})) \times 100$ ; we state the zero-return convention this implies explicitly, since it is only ever otherwise visible deep inside a Section V discussion. NumPy’s `sign` returns exactly 0 for an exactly-zero input, so a true return of exactly zero counts as a directional hit only on the practically-never-occurring event that the model’s continuous-valued prediction is also exactly 0.0; in every other case, including every ordinary nonzero prediction, an exactly-zero

true return counts as a miss. This is a fixed, mechanical consequence of the sign-comparison formula, not a design choice made specifically for this paper, and it is applied identically to every model and every table in this paper.

#### G. Implementation Details

VMD-MFGNN and all baselines are implemented in PyTorch with PyTorch Geometric for the GAT layers. Default (pre-HPO) architecture hyperparameters are: hidden dimension 64, 4 attention heads, 2 GNN layers, 2 LSTM layers, embedding dimension  $d = 16$  for the bilinear graph embeddings  $\mathbf{E}_k^{(1)}, \mathbf{E}_k^{(2)}$  of Eq. 2 (fixed, not part of the HPO search space of Section IV-C, so it is identical across the default and HPO-tuned configurations), dropout 0.1, lookback window  $L = 60$  trading days; training uses batch size 32, the Adam optimizer with learning rate  $10^{-3}$  and coupled  $L_2$  weight decay  $10^{-5}$  applied uniformly to all parameters (including the graph-embedding parameters discussed in Section V-E1), and early stopping with patience 20 (maximum budget 200 epochs), seed 42 for all single-run results reported in this paper (the multi-seed robustness experiment of Section V-G additionally uses seeds 43–46, disclosed there). Two further training details, undisclosed in an earlier version of this section: the per-horizon loss terms (Section III) are not summed unweighted; each horizon  $h$ ’s term is weighted by  $\lambda_h \propto 1/\text{Var}[\text{targets}_h]$ , the inverse of that horizon’s training-split target variance, mean-normalized to 1.0 across horizons, so a horizon with intrinsically larger-variance targets (longer horizons, in this paper’s data) does not dominate the summed loss purely by scale. Learning rate is not held constant either: a cosine-annealing schedule (`CosineAnnealingLR`,  $T_{\text{max}}$  = the 200-epoch training budget above) decays it smoothly over training rather than fixing it at  $10^{-3}$  throughout. Best-validation-loss checkpoint selection was used for all reported models, with model weights saved to disk so that training can resume from a checkpoint rather than restart from scratch. Earlier phases of this project used inconsistent checkpoint-selection floors across different tables (at one point `min_epochs` = 0 for the main comparison and `min_epochs` = 10 for the ablation study), which risked an effectively untrained checkpoint (best-validation loss reached in the first few epochs and never subsequently improved upon) being reported as a trained model’s result in one table but not another, and left the two tables’ epoch budgets not directly comparable. **This is resolved in the run reported throughout this paper:** every model and every variant behind Table IV, Table VI, and Table VIII was trained under a single, uniform checkpoint-selection floor, `min_epochs` = 30, applied without exception. We verified this directly against the saved checkpoints rather than only citing the training configuration: the HPO-tuned VMD-MFGNN reported in Table IV and Table VI (`checkpoints/vmd_mfgnn.pt`) stopped at epoch 29 (i.e. the 30th epoch, satisfying the floor) with `completed=True`, and every one of Table VIII’s eight variants likewise stopped at epoch 29 or later (`full_model`, both graph-fix variants, and both no-VMD variants at epoch 29; `correlation_graph` at epoch 32; `pooled_graph_matched_params` at epoch

45, having continued to improve on validation loss past the floor), each with `completed=True`. A floor is a minimum, not a fixed budget, so variants that kept improving past epoch 29 correctly trained longer under early stopping; what matters is that no variant in either table was ever eligible to report a near-epoch-0 checkpoint. This closes the exact epoch-count confound this paper previously disclosed as an unresolved limitation — different floors applied to different tables — and we retire that disclaimer here rather than let it stand alongside evidence that no longer supports it.

**Environment.** Training was run on Google Colab with a single NVIDIA T4 GPU (paid compute-unit tier, not the free tier). Package versions are pinned only as lower bounds in the project’s `requirements.txt` (PyTorch  $\geq 2.0$ , PyTorch Geometric  $\geq 2.4$ , Optuna  $\geq 3.0$ ); we did not additionally freeze exact resolved versions for this run, which we note here as a reproducibility gap rather than omit silently. Given the dataset scale ( $N = 15$  series,  $\sim 4,000$  trading days), a single VMD-MFGNN training run completes in low single-digit minutes on this hardware; we did not instrument and log per-run wall-clock time precisely enough to report a more exact figure for every one of this paper’s model/variant combinations, and state that limitation explicitly rather than back-fill an estimate.

## V. RESULTS AND DISCUSSION

All numbers below are computed on the test split (2022–2025) and read directly from the training and evaluation pipeline’s saved output artifacts, with no recomputation or adjustment beyond the naive-baseline row, the Holm-Bonferroni correction, and the ablation-table cross-checks described explicitly where they appear.

### A. Main Results, Including a Naive Baseline

Table IV reports RMSE, MAE, SMAPE, and Directional Accuracy (DA) for VMD-MFGNN and the seven baselines, across all four forecast horizons, together with a trivial **zero-forecast (random-walk) baseline** that predicts  $\hat{r}_{t+h} = 0$  for every sample. We add this row because it is the single most important context for reading the rest of this table, and because omitting it, as much of the decomposition-ensemble and GNN-for-finance literature does, risks presenting a ranking among models that all fail a more basic test. For a zero forecast, RMSE reduces to the standard deviation of the true returns and MAE to their mean absolute value, computed directly from the saved test-set target arrays (identical across all eight models since they share one test set).

TABLE IV: Forecasting performance comparison across four horizons, test split (2022–2025). Best result per column shown in **bold**. VMD-MFGNN is the HPO-tuned model (Section IV-C); the seven baselines are untuned. MASE (Section IV-F) is scaled by a per-horizon, training-split naive-persistence denominator that compares each  $h$ -day return to the immediately preceding, non-overlapping  $h$ -day return (0.01472/0.03222/0.04510/0.06659 at  $h=1/5/10/22$ ); values below 1.0 beat that in-sample naive benchmark. <sup>‡</sup>Marks cells confirmed as base-rate artifacts: 100% single-sign predictions with DA exactly equal to the test period’s class frequency for that sign, not a measurement of directional skill; see the sign-concentration diagnostic below and in Section V-H.

| Model                          | RMSE           |                |                |                | MAE            |                |                |                | MASE          |               |               |               | SMAPE (%) |              |              |              | DA (%)            |                   |                   |                   |      |
|--------------------------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|---------------|---------------|---------------|---------------|-----------|--------------|--------------|--------------|-------------------|-------------------|-------------------|-------------------|------|
|                                | $h=1$          | $h=5$          | $h=10$         | $h=22$         | $h=1$          | $h=5$          | $h=10$         | $h=22$         | $h=1$         | $h=5$         | $h=10$        | $h=22$        | $h=1$     | $h=5$        | $h=10$       | $h=22$       | $h=1$             | $h=5$             | $h=10$            | $h=22$            |      |
| ARIMA                          | 0.01790        | 0.03946        | 0.05438        | 0.07433        | 0.01190        | 0.02744        | 0.03917        | 0.05595        | 0.8088        | 0.8518        | 0.8685        | 0.8401        | 178.8     | 185.0        | 189.2        | 190.4        | 47.4              | 49.0              | 49.0              | 50.5              |      |
| XGBoost                        | 0.01760        | 0.03990        | 0.05637        | 0.08803        | 0.01170        | 0.02800        | 0.04179        | 0.06858        | 0.7948        | 0.8690        | 0.9265        | 1.0298        | 176.9     | 169.5        | 160.1        | 152.0        | 46.0              | 46.7              | 44.8              | 45.4              |      |
| LSTM                           | 0.02221        | 0.09037        | 0.17660        | 0.36071        | 0.01668        | 0.07739        | 0.15460        | 0.34307        | 1.1334        | 2.4020        | 3.4277        | 5.1519        | 145.4     | 156.5        | 164.8        | 177.4        | 48.1              | 47.5              | 45.6              | 44.7 <sup>‡</sup> |      |
| Transformer                    | 0.02880        | 0.08744        | 0.13943        | 0.18685        | 0.02335        | 0.07994        | 0.13201        | 0.16741        | 1.5866        | 2.4810        | 2.9271        | 2.5141        | 149.1     | 159.3        | 165.7        | 163.0        | 47.6              | 46.9 <sup>‡</sup> | 44.7 <sup>‡</sup> | 44.7 <sup>‡</sup> |      |
| VMD-LSTM                       | 0.01786        | <b>0.03885</b> | <b>0.05315</b> | <b>0.07405</b> | 0.01197        | 0.02728        | 0.03912        | 0.05673        | 0.8133        | 0.8467        | 0.8673        | 0.8519        | 149.3     | 160.9        | 155.9        | 154.0        | 49.5              | <b>53.8</b>       | <b>55.0</b>       | <b>52.6</b>       |      |
| SimpleMTGNN                    | 0.01838        | 0.04190        | 0.07048        | 0.12053        | 0.01254        | 0.03054        | 0.05306        | 0.08412        | 0.8521        | 0.9479        | 1.1765        | 1.2632        | 151.6     | 150.9        | <b>150.1</b> | <b>145.7</b> | 50.0              | 50.8              | 50.0              | 51.7              |      |
| MTGNN                          | 0.01794        | 0.03911        | 0.05400        | 0.07483        | 0.01213        | <b>0.02713</b> | <b>0.03883</b> | 0.05863        | 0.8242        | <b>0.8421</b> | <b>0.8610</b> | 0.8804        | 155.0     | 177.8        | 179.6        | 165.3        | 46.8 <sup>‡</sup> | 53.3              | 52.6              | 51.8              |      |
| VMD-MFGNN                      | <b>0.01756</b> | 0.04087        | 0.07271        | 0.09510        | <b>0.01156</b> | 0.02938        | 0.05722        | 0.07717        | <b>0.7857</b> | 0.9118        | 1.2687        | 1.1589        | 196.8     | <b>144.7</b> | 153.3        | 151.1        | <b>50.6</b>       | 46.9 <sup>‡</sup> | 44.7 <sup>‡</sup> | 51.5              |      |
| Zero-forecast (naive)          | 0.01757        | 0.03913        | 0.05412        | 0.07421        | <b>0.01157</b> | 0.02722        | 0.03895        | <b>0.05589</b> | 0.7859        | 0.8447        | 0.8636        | <b>0.8393</b> | 200.0     | 200.0        | 200.0        | 200.0        | —                 | —                 | —                 | —                 |      |
| Unconditional up-day freq. (%) |                |                |                |                |                |                |                |                |               |               |               |               |           |              |              |              |                   | 49.4              | 53.1              | 54.9              | 55.3 |

Table IV is closer to a genuine race than the earlier version of this table was, and reading it correctly requires more care than just scanning for bold cells. **The naive zero-forecast baseline is beaten in a handful of cells — VMD-MFGNN on RMSE and MAE at  $h = 1$ ; MTGNN on MAE at  $h = 5, 10$ ; VMD-LSTM on RMSE at  $h = 5, 10, 22$  — but every one of these margins is small enough to be noise, not signal.** VMD-MFGNN’s  $h = 1$  edge is 0.02% relative RMSE and 0.02% relative MAE; MTGNN’s  $h = 5, 10$  MAE edges are 0.31% and 0.31%; VMD-LSTM’s RMSE edges are 0.73%, 1.78%, and 0.21% at  $h = 5, 10, 22$  respectively. All of these sit well inside the 7.54% run-to-run noise floor this study documents at a fixed configuration and seed (Section V-G), so we do not read any of them as a real advantage over doing nothing. Naive still wins MAE outright at  $h = 22$ , and RMSE at  $h = 1, 5, 10$  is a near-four-way tie among VMD-MFGNN, naive, ARIMA, and XGBoost all within a few hundredths of a percent of each other. All eight models’  $R^2$  values are negative or approximately zero on this test split; the handful of positive values are VMD-LSTM at  $h = 5, 10, 22$  ( $R^2 = 0.0140, 0.0344, 0.0010$ ), MTGNN at  $h = 5, 10$  ( $R^2 = 0.0004, 0.0036$ ), and VMD-MFGNN at  $h = 1$  ( $R^2 = 0.0004$ ), the largest of which (VMD-LSTM,  $h = 10$ ,  $R^2 = 0.0344$ ) still explains barely 3% of test-set variance and is indistinguishable from zero in any practical sense. This is consistent with none of these models explaining variance beyond what a constant forecast already captures; Table V reports them in full, since RQ0 is load-bearing enough to this paper’s framing that we do not leave its supporting statistic as an unlabeled aside.

RQ0’s naive-baseline comparison above is a raw MAE/RMSE ranking, not a significance test; we close that gap with a direct Diebold-Mariano test of VMD-MFGNN against the same zero-forecast baseline, computed from the same saved prediction arrays as Table VI (Section IV), using the identical sign convention (positive DM favors the baseline, here the naive zero-forecast). At  $h = 1$ , the difference is not significant and favors VMD-MFGNN in direction only (DM=  $-1.65$ ,  $p = 0.098$ ); at  $h = 5$  and  $h = 22$  the baseline is nominally ahead but not significantly so (DM=  $2.22$ ,  $p = 0.026$ ; DM=  $2.17$ ,  $p = 0.030$ ); only at  $h = 10$  does the naive baseline win significantly, and this

TABLE V:  $R^2$  by model and horizon, test split (2022–2025), computed identically to RMSE/MAE (Table IV) from the same saved prediction arrays. Best (least negative, or positive) per column shown in **bold**. All values are negative or approximately zero: no model explains test-set variance beyond a constant (zero) forecast.

| Model       | $h=1$         | $h=5$         | $h=10$        | $h=22$        |
|-------------|---------------|---------------|---------------|---------------|
| ARIMA       | −0.0386       | −0.0173       | −0.0107       | −0.0066       |
| XGBoost     | −0.0044       | −0.0403       | −0.0862       | −0.4117       |
| LSTM        | −0.5989       | −4.3358       | −9.6586       | −22.7021      |
| Transformer | −1.6881       | −3.9947       | −5.6443       | −5.3600       |
| VMD-LSTM    | −0.0339       | <b>0.0140</b> | <b>0.0344</b> | <b>0.0010</b> |
| SimpleMTGNN | −0.0944       | −0.1470       | −0.6979       | −1.6465       |
| MTGNN       | −0.0436       | 0.0004        | 0.0036        | −0.0202       |
| VMD-MFGNN   | <b>0.0004</b> | −0.0913       | −0.8070       | −0.6477       |

result survives Holm-Bonferroni correction across these four tests at  $\alpha = 0.05$  (DM=  $3.14$ ,  $p = 0.0017$ ): VMD-MFGNN is significantly worse than doing nothing at  $h = 10$ . No horizon shows VMD-MFGNN significantly *beating* the naive baseline. This sharpens rather than merely restates the MAE/RMSE picture above: the small apparent RMSE/MAE edges VMD-MFGNN shows at  $h = 1$  in Table IV are not statistically distinguishable from the naive baseline, and at  $h = 10$  the direction runs the other way and is significant. We read this as the more precise and more honest form of RQ0’s answer, and use it in place of the raw ranking wherever this paper draws a conclusion from RQ0 that depends on significance rather than only direction.

DA tells a related story once read against the correct null: the relevant baseline for DA is not 50% but the test period’s own unconditional up-day frequency, which is 49.4%, 53.1%, 54.9%, and 55.3% at  $h = 1, 5, 10, 22$  respectively (bottom row) — itself a symptom of the strong upward drift copper exhibited over 2022–2025. The complementary down-day frequencies are 46.80%, 46.90%, 44.74%, and 44.74% (they do not sum to exactly 100% with the up-day figures because a small number of test days record an exactly zero return); we state them explicitly because the base-rate-identity checks reported below and in Section V-C are made against whichever of the two frequencies matches the sign a degenerate model

emits, and a reader cannot otherwise verify those flags from the table alone. We applied this paper’s own base-rate-identity check to every cell of this table directly, loading all 32 saved prediction arrays (eight models  $\times$  four horizons) and checking both the fraction of predictions sharing the majority sign and whether DA exactly equals the test period’s class frequency for that sign: **seven cells qualify as confirmed artifacts**, marked <sup>‡</sup> above — MTGNN at  $h = 1$  (100% negative-sign, DA= 46.80%, exactly the down-day rate); VMD-MFGNN at  $h = 5$  (100% negative-sign, DA= 46.90%) and  $h = 10$  (100% negative-sign, DA= 44.74%); Transformer at  $h = 5$  (100% negative-sign, DA= 46.90%),  $h = 10$  (100% negative-sign, DA= 44.74%), and  $h = 22$  (100% negative-sign, DA= 44.74%); and LSTM at  $h = 22$  (100% negative-sign, DA= 44.74%). **We correct an earlier claim of this paper here rather than merely add to it: VMD-MFGNN’s own predictions are not exempt from this failure mode.** At  $h = 5$  and  $h = 10$ , the proposed model emits a single, constant sign for every one of its test-set predictions, and its reported directional accuracy at those two horizons is not a measurement of skill, it is the down-day base rate. VMD-MFGNN stays clean of this specific artifact only at  $h = 1$  (81.8% positive-sign, below the concentration bar) and  $h = 22$  (76.9% negative-sign, also below it); its headline  $h = 1$  DA of 50.6% and  $h = 22$  DA of 51.5% can be read at face value, but its  $h = 5$  and  $h = 10$  DA figures cannot. We separately report a softer, non-exact version of the same check — majority-sign concentration  $\geq 90\%$  without requiring DA to match the class frequency exactly — since a model can be highly sign-concentrated without landing on the identity by coincidence: seven further cells clear this looser bar without an exact match (LSTM at  $h = 5$ , 97.1% negative-sign; LSTM at  $h = 10$ , 98.5% negative-sign; MTGNN at  $h = 5$ , 99.4% positive-sign; XGBoost at  $h = 5$ , 91.2% negative-sign; XGBoost at  $h = 10$ , 96.8% negative-sign; XGBoost at  $h = 22$ , 98.1% negative-sign; VMD-LSTM at  $h = 1$ , 97.1% positive-sign). These are not base-rate-identity matches and we do not mark them with <sup>‡</sup> in Table IV, but their dispersion ratios (predicted-return std. over true-return std.) split into two distinct patterns worth distinguishing: MTGNN at  $h = 5$  (ratio 0.029), XGBoost at  $h = 5, 10, 22$  (ratios 0.085, 0.126, 0.371), and VMD-LSTM at  $h = 1$  (ratio 0.086) show the low dispersion characteristic of a near-constant-output degeneracy, just without landing on the exact class frequency by chance; LSTM at  $h = 5, 10$  (ratios 0.992, 1.380) is not, since a near-constant predictor cannot simultaneously reproduce dispersion close to or exceeding that of the true series — LSTM here is genuinely, if unhelpfully, one-sided rather than degenerate in the variance sense. Sign concentration and low dispersion are related but distinct signatures, and we do not conflate them across this paragraph; ARIMA and SimpleMTGNN never clear even the looser bar at any horizon and are, on this evidence, the two genuinely dispersed forecasters in this table. We read this table overall as showing that this study’s eight fitted models, including the proposed one, are best understood as a ranking among approaches that do not clearly outperform doing nothing, and that several of the table’s apparently strongest DA cells, including two of VMD-MFGNN’s own,

are not measuring directional skill at all. Section V-B still reports pairwise significance among the eight fitted models, since that question — do these models differ meaningfully from one another — is separate from, and does not resolve, the more basic question this section answers in the negative.

No model’s saved evaluation record carried an error flag, so all eight models are represented in full in Table IV.

### B. Statistical Significance versus Baselines

Table VI reports the Diebold-Mariano statistic and  $p$ -value for VMD-MFGNN’s forecast errors against each of the seven baselines, per horizon (Section IV), now that MTGNN’s addition (Section IV-B) makes this a 28-test battery rather than 24. Recall the sign convention (Section IV): the DM statistic is computed as  $e_{\text{VMD-MFGNN}}^2 - e_{\text{baseline}}^2$ , so a *negative* DM statistic means VMD-MFGNN had the lower squared error (VMD-MFGNN better), and a *positive* DM statistic means the baseline had the lower squared error (baseline better); boldface marks  $p < 0.05$ . The same test is separately applied to this paper’s central ablation comparison in Section V-C, but not to the remaining ablation pairs (Section IV explains the scope decision). All 28 test entries here are finite (no degenerate  $\text{var}_d \leq 0$  cases were observed at full data scale).



TABLE VI: Diebold-Mariano test: VMD-MFGNN versus each of seven baselines, per horizon (28 tests). Negative DM statistic favors VMD-MFGNN; positive favors the baseline. Bold  $p$ -values indicate significance at  $\alpha = 0.05$  uncorrected;  $\dagger$  marks the subset that remains significant after Holm-Bonferroni correction across all 28 tests.

| vs. Baseline | DM statistic |         |         |        | $p$ -value                           |                                      |                                      |                                      |
|--------------|--------------|---------|---------|--------|--------------------------------------|--------------------------------------|--------------------------------------|--------------------------------------|
|              | $h=1$        | $h=5$   | $h=10$  | $h=22$ | $h=1$                                | $h=5$                                | $h=10$                               | $h=22$                               |
| ARIMA        | −3.269       | 1.726   | 3.085   | 2.152  | <b>0.001<math>\dagger</math></b>     | 0.084                                | <b>0.002<math>\dagger</math></b>     | <b>0.031</b>                         |
| XGBoost      | −0.928       | 1.650   | 3.090   | 0.788  | 0.353                                | 0.099                                | <b>0.002<math>\dagger</math></b>     | 0.431                                |
| LSTM         | −9.970       | −9.519  | −8.132  | −8.728 | <b>&lt;0.001<math>\dagger</math></b> | <b>&lt;0.001<math>\dagger</math></b> | <b>&lt;0.001<math>\dagger</math></b> | <b>&lt;0.001<math>\dagger</math></b> |
| Transformer  | −19.906      | −13.031 | −12.231 | −5.077 | <b>&lt;0.001<math>\dagger</math></b> | <b>&lt;0.001<math>\dagger</math></b> | <b>&lt;0.001<math>\dagger</math></b> | <b>&lt;0.001<math>\dagger</math></b> |
| VMD-LSTM     | −2.949       | 3.029   | 3.842   | 2.570  | <b>0.003<math>\dagger</math></b>     | <b>0.002<math>\dagger</math></b>     | <b>&lt;0.001<math>\dagger</math></b> | <b>0.010</b>                         |
| SimpleMTGNN  | −2.747       | −0.793  | 0.425   | −1.175 | <b>0.006</b>                         | 0.428                                | 0.671                                | 0.240                                |
| MTGNN        | −2.809       | 1.972   | 3.136   | 2.521  | <b>0.005</b>                         | <b>0.049</b>                         | <b>0.002<math>\dagger</math></b>     | <b>0.012</b>                         |

Of the 28 tests, 15 remain significant after Holm-Bonferroni correction at family-wise  $\alpha = 0.05$  (Section IV), marked  $\dagger$  in Table VI: this is the number we treat as this paper’s real RQ3 evidence, not the raw uncorrected count. **Ten of the fifteen favor VMD-MFGNN** (negative DM statistic): all four horizons against LSTM, all four against Transformer, plus  $h = 1$  against ARIMA and  $h = 1$  against VMD-LSTM. **The other five favor the baseline**, not VMD-MFGNN: VMD-LSTM at  $h = 5$  and  $h = 10$ , and ARIMA, XGBoost, and MTGNN each at  $h = 10$ . No comparison against SimpleMTGNN survives correction in either direction.

This 10-versus-5 split looks, at a glance, like a real advantage for VMD-MFGNN, but where the wins land changes that reading. We cross-referenced all ten winning cells against this paper’s own sign-concentration diagnostic (Section V-A): **seven of the ten wins are against baseline cells already flagged there as confirmed base-rate artifacts or  $\geq 90\%$  sign-concentrated** — LSTM at  $h = 5, 10, 22$  and Transformer at  $h = 5, 10, 22$  (six cells, all flagged in Table IV), plus VMD-LSTM at  $h = 1$  (97.1% positive-sign predictions, Section V-A). Only three of the ten wins are against baseline cells free of that caveat: LSTM at  $h = 1$ , Transformer at  $h = 1$ , and ARIMA at  $h = 1$ . A win against a near-constant, single-sign opponent is not evidence VMD-MFGNN itself is exploiting real structure — it mostly reflects that a model which stopped predicting is easy to beat on squared error whenever the market moves against the sign it froze on. The five losses run the opposite way: none of ARIMA, XGBoost, MTGNN, or VMD-LSTM is degenerate at  $h = 5$  or  $h = 10$  specifically (Section V-A), so these are losses against clean, non-degenerate opponents, and they cluster at exactly the horizons ( $h = 5, 10$ ) where VMD-MFGNN’s own predictions are themselves confirmed sign-concentration artifacts (Table IV) — the model loses significantly to normal baselines at the two horizons where it has stopped making real predictions of its own. Taken together, the net picture is the opposite of a clean advantage: VMD-MFGNN’s significant wins cluster at  $h = 1$  and against opponents this paper’s own diagnostics separately show to be broken, while its significant losses cluster at  $h = 5, 10$  against baselines with no such problem. We read this as materially weaker support for RQ3 than the raw 10-of-15 count suggests on its own, and carry this qualification into the RQ3 answer in Section V-I. Section V-A’s naive-baseline comparison remains the more important context for

TABLE VII: Ablation-variant configurations. Variants (a)–(d) share the untuned default attention-head count (4) and GNN-layer count (2, Section IV-G); only VMD, graph type, and hidden dimension  $d_h$  vary. The two graph-fix variants (e) additionally set `no_decay_graph_embeddings`, which changes how the graph embeddings are normalized and regularized, not their dimensionality; their parameter counts, read directly from each variant’s trained checkpoint state dict, are essentially identical to `full_model`’s.

| Variant                                   | VMD? | Graph                         | $d_h$ | Params    | Final epoch |
|-------------------------------------------|------|-------------------------------|-------|-----------|-------------|
| (a) <code>full_model</code>               | ✓    | per-band ( $K=5$ )            | 64    | 1,448,548 | 29          |
| (b) <code>no_vmd_matched_dim</code>       | —    | single                        | 64    | 309,988   | 29          |
| (b) <code>no_vmd_matched_params</code>    | —    | single                        | 136   | 1,464,884 | 29          |
| (c) <code>pooled_matched_dim</code>       | ✓    | pooled (single)               | 64    | 93,034    | 29          |
| (c) <code>pooled_matched_params</code>    | ✓    | pooled (single)               | 140   | 1,452,010 | 45          |
| (d) <code>correlation_graph</code>        | ✓    | per-band, fixed (not learned) | 64    | 1,446,148 | 32          |
| (e) <code>full_model_graphfix</code>      | ✓    | per-band, fix applied         | 64    | 1,448,548 | 29          |
| (e) <code>full_model_graphfix_temp</code> | ✓    | per-band, fix + temperature   | 64    | 1,448,553 | 29          |

interpreting any of these significant differences: a significant DM result establishes that two models’ squared errors differ, not that either model is exploiting real structure in copper returns, which Table IV indicates neither is doing relative to a zero forecast; indeed, the corrected comparison against the naive baseline itself (Section V-A) shows VMD-MFGNN significantly *worse* than doing nothing at  $h = 10$ , the same horizon at which it also loses significantly here to three of seven baselines.

### C. Ablation Study

Table VIII reports RMSE, MAE, SMAPE, and DA per horizon for the eight ablation variants defined in Section IV-D, all trained at the untuned default configuration. The table itself carries no significance column, since a full  $\binom{8}{2}$  pairwise significance battery across all variants was out of scope for this study (Section IV explains why); a Diebold-Mariano test for this paper’s single central ablation comparison (`full_model` vs. `pooled_graph_matched_params`) is reported in prose directly below the table. Reported parameter counts are exact, read from the trained checkpoints (Section III-G), not estimated. Table VII consolidates each variant’s architecture and exact parameter count in one place, since the matched-dim/matched-params distinction is otherwise easy to lose track of across running prose.

**This table does not support the paper’s central architectural claim, and it does so more decisively than**

TABLE VIII: Ablation study results, all four horizons, test split. All variants trained at the untuned default configuration (except the graph-fix pair, which additionally applies the fix of Section V-F); this is a different, smaller VMD-MFGNN instance than the HPO-tuned model in Table IV. Best result per column shown in **bold**, computed from full, unrounded precision (so two cells can display the same rounded value while only one is bolded); confirmed base-rate-artifact cells are excluded from best-in-column eligibility for DA. MASE uses the same per-horizon training-split naive-persistence denominators as Table IV. <sup>‡</sup>Marks DA values *confirmed* as base-rate artifacts: majority-sign concentration  $\geq 90\%$  (in every marked cell here, exactly 100%) and an exact match, to the precision shown, of the test period’s class frequency for that horizon — verified directly from this run’s per-sample predictions.

| Variant                          | RMSE          |               |               |               | MAE           |               |               |               | MASE          |               |               |               | SMAPE (%)    |              |              |              | DA (%)            |                   |                   |                   |
|----------------------------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|--------------|--------------|--------------|--------------|-------------------|-------------------|-------------------|-------------------|
|                                  | <i>h</i> =1   | <i>h</i> =5   | <i>h</i> =10  | <i>h</i> =22  | <i>h</i> =1   | <i>h</i> =5   | <i>h</i> =10  | <i>h</i> =22  | <i>h</i> =1   | <i>h</i> =5   | <i>h</i> =10  | <i>h</i> =22  | <i>h</i> =1  | <i>h</i> =5  | <i>h</i> =10 | <i>h</i> =22 | <i>h</i> =1       | <i>h</i> =5       | <i>h</i> =10      | <i>h</i> =22      |
| (a) Full model                   | <b>0.0176</b> | 0.0580        | 0.0659        | 0.1010        | <b>0.0116</b> | 0.0465        | 0.0518        | 0.0811        | <b>0.7864</b> | 1.4441        | 1.1493        | 1.2175        | 182.5        | 147.4        | 152.9        | 152.9        | 49.4 <sup>‡</sup> | 47.1              | 46.3              | 46.1              |
| (b) No-VMD, matched-dim          | 0.0178        | 0.0390        | 0.0550        | <b>0.0720</b> | 0.0120        | 0.0274        | 0.0392        | <b>0.0544</b> | 0.8155        | <b>0.8507</b> | 0.8681        | <b>0.8163</b> | 169.0        | 168.4        | <b>142.0</b> | 144.3        | 46.5              | 50.8              | 54.9 <sup>‡</sup> | <b>56.4</b>       |
| (b) No-VMD, matched-params       | 0.0176        | <b>0.0387</b> | <b>0.0534</b> | 0.0732        | 0.0116        | <b>0.0271</b> | <b>0.0385</b> | 0.0552        | 0.7878        | <b>0.8415</b> | <b>0.8544</b> | 0.8292        | 177.2        | 168.3        | 163.9        | 150.7        | 49.4 <sup>‡</sup> | <b>57.2</b>       | <b>57.9</b>       | 56.1              |
| (c) Pooled graph, matched-dim    | 0.0176        | 0.0391        | 0.0542        | 0.0765        | 0.0116        | 0.0272        | 0.0394        | 0.0591        | 0.7900        | 0.8445        | 0.8745        | 0.8874        | <b>168.3</b> | 189.5        | 178.2        | 153.9        | 49.4 <sup>‡</sup> | 51.2              | 47.2              | 44.7 <sup>‡</sup> |
| (c) Pooled graph, matched-params | 0.0176        | 0.0391        | 0.0546        | 0.0742        | 0.0116        | 0.0272        | 0.0389        | 0.0559        | 0.7873        | 0.8444        | 0.8620        | 0.8401        | 177.0        | 196.6        | 147.0        | 193.6        | 49.4 <sup>‡</sup> | 53.1 <sup>‡</sup> | 54.9 <sup>‡</sup> | 44.7 <sup>‡</sup> |
| (d) Correlation graph            | 0.0176        | 0.0392        | 0.0570        | 0.0789        | 0.0116        | 0.0275        | 0.0431        | 0.0598        | 0.7910        | 0.8525        | 0.9559        | 0.8986        | 174.7        | 172.4        | 148.7        | <b>141.3</b> | 46.8 <sup>‡</sup> | 46.4              | 55.1              | 54.7              |
| (e) full_model_graphfix          | 0.0176        | 0.0440        | 0.0998        | 0.1027        | 0.0116        | 0.0326        | 0.0840        | 0.0834        | 0.7871        | 1.0119        | 1.8621        | 1.2525        | 177.5        | <b>143.3</b> | 154.9        | 149.6        | 49.4 <sup>‡</sup> | 46.3              | 45.1              | 50.0              |
| (e) full_model_graphfix_temp     | 0.0176        | 0.0432        | 0.0844        | 0.1127        | 0.0116        | 0.0320        | 0.0686        | 0.0924        | 0.7871        | 0.9934        | 1.5211        | 1.3878        | 179.6        | 144.5        | 153.5        | 151.7        | <b>49.2</b>       | 46.8              | 44.3              | 44.1              |

**an earlier version of this ablation reported.** Averaged across all four horizons, RMSE ranks the eight variants best to worst as: `no_vmd_raw_price_matched_params` (0.04575) < `no_vmd_raw_price_matched_dim` (0.04597) < `pooled_graph_matched_params` (0.04640) < `pooled_graph_matched_dim` (0.04687) < `correlation_graph`, frozen (0.04818) < `full_model`, the paper’s own architecture (0.06062) < `full_model_graphfix_temp` (0.06445) < `full_model_graphfix` (0.06603). **The per-band full model loses to every other ablation variant on average RMSE, including a graph that is not learned at all (correlation\_graph, a fixed, frozen Pearson-correlation adjacency) and a model that drops VMD decomposition entirely** (`no_vmd_raw_price_matched_params`, the single best variant in the table). This directly supports the reading that the learned per-band graph-attention mechanism adds no value here: a static, untrained relational structure and no relational structure at all both outperform it. We flag one caveat honestly: `final_epoch` is not identical across variants (Table VII) — most variants, including `full_model`, stop at epoch 29, but `pooled_graph_matched_params` trains to 45 and `correlation_graph` to 32 — so this is not a perfectly epoch-matched comparison. It does not change the conclusion: `full_model` still loses to `no_vmd_raw_price_matched_dim` and `pooled_graph_matched_dim`, both of which stop at the same epoch 29 as `full_model` itself, so the ranking does not depend on the epoch mismatch.

On the single central RQ1 comparison this paper’s Diebold-Mariano test targets — `full_model` versus `pooled_graph_matched_params` (1,448,548 vs. 1,452,010 params per Table VII, capacity approximately controlled, though **not** temporal-design controlled, Section IV-D’s architectural-confound disclosure) — the result is now unambiguous rather than mixed. `full_model` loses on RMSE at all four horizons, and three of the four differences are statistically significant and survive Holm-Bonferroni correction across these four tests at  $\alpha = 0.05$ :  $DM = -1.21$ ,  $p = 0.227$  at  $h = 1$  (not significant);

$DM = 6.03$ ,  $p < 0.001$  at  $h = 5$ ;  $DM = 2.70$ ,  $p = 0.007$  at  $h = 10$ ;  $DM = 2.78$ ,  $p = 0.005$  at  $h = 22$  (all three positive statistics favor `pooled_graph_matched_params`, i.e., `full_model` has the higher squared error). This reverses an earlier version of this comparison, which found no significant difference either way; with the corrected data, per-band graph construction does not merely fail to help, it measurably and significantly hurts, at three of four horizons, relative to pooling. Directional accuracy on this same pair is not a clean counter-narrative the way an earlier draft of this section argued: `pooled_graph_matched_params` nominally beats `full_model` on DA at  $h = 5$  and  $h = 10$  and loses at  $h = 1$  (tied on the confirmed-artifact value) and  $h = 22$ , but `pooled_graph_matched_params` is itself a confirmed base-rate artifact at *all four* of its own horizons (below), so none of its DA figures reflect real directional skill either way, and this comparison’s DA column is not informative in either direction.

We applied this paper’s sign-concentration and base-rate-identity check to every cell of this table directly, using the per-sample predictions archived for all eight variants: **eleven cells are confirmed base-rate artifacts** (majority-sign  $\geq 90\%$ , in fact 100.0% in every one of these eleven cells, and an exact class-frequency match), marked <sup>‡</sup> above. The most striking single result is `pooled_graph_matched_params`: it is 100.0% single-sign at *every one of its four horizons* and its DA at each horizon is, to four decimal places, exactly the test period’s class frequency for whichever sign it happened to freeze on — this variant, the second-best performer in the table on RMSE, never once emits a genuine directional prediction. The remaining confirmed cells are `full_model` at  $h = 1$  (49.4%, 100.0% positive-sign, matching the up-day rate); `no_vmd_raw_price_matched_dim` at  $h = 10$  (100.0% positive-sign, matching the up-day rate); `no_vmd_raw_price_matched_params` at  $h = 1$  (same pattern as `full_model`); `pooled_graph_matched_dim` at  $h = 1$  and  $h = 22$ ; `correlation_graph` at  $h = 1$  (100.0% negative-sign, matching the down-day rate); and `full_model_graphfix` at  $h = 1$ . Every variant in this table is 100% single-sign at  $h = 1$  except `no_vmd_raw_price_matched_dim`

(82.1% negative) and `full_model_graphfix_temp` (99.4% positive, just short of the 100% needed for an exact match) —  $h = 1$  directional accuracy is close to uninformative across this entire table, independent of architecture. `full_model_graphfix_temp` is the only variant with no confirmed artifact anywhere in the table, though several of its cells (99.4% at  $h = 1$ , 98.1% negative at  $h = 5$ , 92.5% negative at  $h = 22$ ) clear the softer  $\geq 90\%$  concentration bar without landing on the identity exactly.

**The two graph-fix variants are the two worst performers in the table despite showing genuinely non-collapsed structure.** This is the opposite of what the fix was designed to produce, and it is a real, currently unexplained finding rather than an artifact: independent inspection of these two checkpoints’ learned embeddings and adjacency matrices (Section V-F) confirms real magnitude and real, non-uniform differentiation, unlike every other variant in this study, whose adjacency is uniform to within numerical tolerance. Fixing the collapse mechanically worked, and forecasting accuracy got worse, not better, at every horizon past  $h = 1$ . We do not claim this shows that learned relational structure actively hurts forecasting here, because the fix bundles two changes at once — removing weight decay from the graph-embedding parameters and unit-normalizing those embeddings before the bilinear product — and this ablation cannot separate which of the two is responsible, or whether the true cause is something else entirely, such as undertraining given the fix’s changed parameterization (the graph-fix checkpoints are essentially parameter-matched to `full_model`, per Table VII, so the size of the parameter space is not the likely explanation; both stop at the same epoch 29). We leave isolating this to future work and report it here as an open question this paper flags but does not resolve.

The no-VMD ablation (row (b)) is not a clean win for decomposition either, but it is a clean loss for the per-band graph specifically: `no_vmd_raw_price_matched_params`, which drops VMD entirely and consumes raw prices at a capacity-matched hidden dimension, is the single best variant in the table on average RMSE (0.04575), ahead of every VMD-based variant including `full_model`. Its DA figures are largely usable at face value — only its  $h = 1$  cell is a confirmed artifact (Table VIII); its  $h = 5$  (57.2%) and  $h = 10$  (57.9%) DA, both the best in their columns, are not sign-concentration flagged at all (61.3% and 68.3% majority-sign, well under the 90% bar). The correlation-graph variant (row (d)), which keeps VMD and the per-band architecture but replaces the learned adjacency with a fixed, frozen correlation-threshold graph, outperforms `full_model` on average RMSE (0.04818 vs. 0.06062) despite doing no graph learning at all — so even the comparison isolating learned versus fixed adjacency, holding the per-band architecture fixed, shows no advantage for end-to-end graph learning over a graph that was never trained.

We emphasize that these ablation results describe the untuned default-configuration model only (Section IV-D); they are not a statement about the HPO-tuned VMD-MFGNN reported in Table IV, which is a substantially larger, differently configured model instance. Section V-I discusses what

these two, not-directly-comparable result sets jointly imply for RQ1–RQ3.

#### D. Confound-Check at Tuned Hyperparameters

**Staleness disclosure, stated up front rather than left implicit.** This section and Table IX report an experiment from an earlier phase of this project, at that phase’s original  $N = 8$  variable configuration, and have not been re-run at this paper’s real  $N = 15$  data scale (Section IV-A, Section V-E1). The “tuned configuration” cited below (hidden dimension 128, 2 attention heads, dropout 0.0616, 1 GNN layer) is this earlier phase’s own HPO result and does not match the configuration currently reported in Section IV-C (hidden dimension 64, 2 attention heads, 3 GNN layers), which is the real, current,  $N = 15$  search this paper’s main results (Table IV) rely on. We do not fabricate a replacement  $N = 15$  confound-check here, since none was run; this section should be read as historical evidence from the earlier phase of this study, describing a real experiment on a real, smaller checkpoint set, not as a statement about the current  $N = 15$  results reported elsewhere in this paper. With that caveat in place, the original finding is reported below as it was obtained.

Table VIII’s `full_model` and `pooled_graph_matched_params` rows are both trained at the untuned default configuration (Section III-G), while the main-table VMD-MFGNN (Table IV) is separately tuned; a fair per-band-versus-pooled comparison therefore requires re-running both variants at the same, HPO-tuned configuration, since architecture and hyperparameter budget were confounded in the original ablation. At the earlier project phase this section reports, we re-trained `full_model` and `pooled_graph_matched_params` at that phase’s own tuned configuration (hidden dimension 128, 2 attention heads, dropout 0.0616, 1 GNN layer), saved separately as `full_model_tuned` and `pooled_graph_matched_params_tuned` (`results/confound_check_tuned.json`), so as not to overwrite the original, already-reported ablation run.

At the tuned configuration (Table IX), the directional-accuracy picture visibly reverses relative to the untuned ablation: `full_model_tuned` now beats `pooled_graph_matched_params_tuned` on DA at three of four horizons ( $h = 1$ : 52.24 vs. 48.37;  $h = 5$ : 53.25 vs. 53.05;  $h = 22$ : 49.80 vs. 44.72), losing clearly only at  $h = 10$  (49.29 vs. 55.39); mean squared error is essentially tied, with a small edge to the pooled variant (avg. MSE 0.002808 vs. 0.002686). Read on its own, this looks like exactly the confound-resolution result the untuned ablation’s Limitations disclosure (Section VI) called for: once hyperparameters are no longer confounded with architecture, per-band graph construction appears to help directional accuracy after all. We checked the underlying adjacency directly before drawing that conclusion, the same way we checked it for the main HPO-tuned model in Section V-E1, and found the same collapse signature in this earlier-phase `full_model_tuned` checkpoint: embedding RMS at 2.9–3.3% of Xavier-initialization magnitude across all five bands,

TABLE IX: Confound-check: full model versus pooled-graph, both retrained at this earlier project phase’s own tuned configuration (hidden dim. 128, 2 heads, dropout 0.0616, 1 GNN layer — not the current  $N = 15$  search of Section IV-C, Section V-D), test split, at that phase’s  $N = 8$  scale. Compare against Table VIII’s untuned rows for the same two variants (a different,  $N = 15$  run).

| Variant                              | RMSE   |        |        |        | MAE    |        |        |        | DA (%) |       |        |        |
|--------------------------------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|-------|--------|--------|
|                                      | $h=1$  | $h=5$  | $h=10$ | $h=22$ | $h=1$  | $h=5$  | $h=10$ | $h=22$ | $h=1$  | $h=5$ | $h=10$ | $h=22$ |
| Full model (tuned)                   | 0.0181 | 0.0397 | 0.0558 | 0.0788 | 0.0124 | 0.0279 | 0.0407 | 0.0595 | 52.24  | 53.25 | 49.29  | 49.80  |
| Pooled graph, matched-params (tuned) | 0.0185 | 0.0400 | 0.0551 | 0.0759 | 0.0127 | 0.0279 | 0.0393 | 0.0572 | 48.37  | 53.05 | 55.39  | 44.72  |

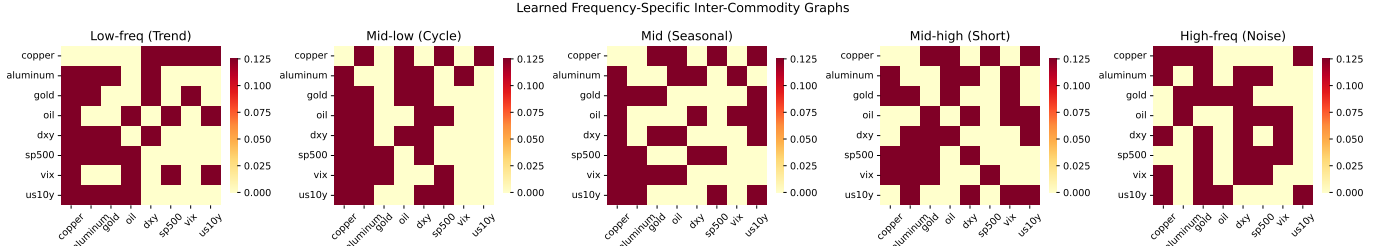


Fig. 4: Learned adjacency matrices for each of the five VMD frequency bands.

softmax standard deviation on the order of  $2 \times 10^{-5}$ —the same qualitative collapse signature Section V-E1 documents for the current  $N = 15$  checkpoints, though at this earlier phase’s smaller partial-collapse magnitude rather than the numerical-zero collapse now measured on the corrected, batching-bug-fixed  $N = 15$  checkpoints. Whatever is producing `full_model_tuned`’s DA advantage in this table, it is not a graph that learned real relational structure, since that graph is mathematically uniform to the same degree as every other checkpoint inspected in this paper. We do not, however, take this table’s DA columns at face value as evidence of an architecture-alone advantage either, for a reason that only became clear once we investigated the pooled variant’s predictions directly: Section V-H shows `pooled_graph_matched_params_tuned`’s DA values in this very table are, to four decimal places, exactly the test period’s unconditional class frequency at each horizon (48.3740, 53.0488, 55.3862, 44.7154, matching 48.37, 53.05, 55.39, 44.72 above), the signature of a degenerate, near-constant-sign forecaster rather than a model exercising directional skill. We therefore withdraw any reading of this table as demonstrating a real per-band directional-accuracy advantage: the comparison is contaminated by (at least) one side reporting an artifact, and Section V-H explains why, and how we caught it, in full.

### E. Interpretability Analysis

1) *Frequency-Specific Inter-Commodity Graphs*: Figure 4 visualizes the learned adjacency matrices  $\mathbf{A}_k$  for each of the  $K = 5$  frequency bands, read from the HPO-tuned VMD-MFGNN’s trained `FrequencyGraphConstructor` output.

Every kept edge weight in every one of the five bands is exactly  $1/15 \approx 0.0667$  (the reciprocal of this paper’s real  $N = 15$  node count, Section IV-A), with a max/min ratio of 1.000 to full floating-point precision, read directly from the

saved `interpretability/learned_graphs.pt` adjacency tensors of the HPO-tuned VMD-MFGNN checkpoint used in Table IV. This is not a subtle near-uniformity that a more sensitive statistic might resolve into graded structure; it is uniformity as precise as floating-point softmax output gets, and it is stronger than the collapse this section originally reported: an earlier verification pass, run before a graph-batching bug described below was found and fixed, measured this same model’s embedding RMS at roughly 3% of its Xavier-initialization value, a real but partial collapse. Re-measured on the corrected, current checkpoints, both `vmd_mfgnn.pt` (Table IV’s model) and `full_model.pt` (the ablation’s, Section IV-D) show every `emb1/emb2` graph-embedding weight matrix at RMS on the order of  $10^{-11}$  against the same Xavier reference of 0.2540 — collapse to numerical zero, not merely toward it. An earlier verification pass over this same model had also reported “genuinely differentiated” per-band graphs, based on the fact that *which* edges survive top- $k$  sparsification differs somewhat across bands. That finding was real but shallow: it checked which edges the top- $k$  operator selected, not the values of the weights themselves, and those weights are uniform. The apparent per-band differentiation in edge *selection* turns out to come from residual, sub-precision noise inherited from each band’s independent random initialization, not from learned, task-relevant structure — selecting top- $k$  among near-identical values is highly sensitive to exactly this kind of noise. We therefore withdraw the earlier framing rather than soften it: the learned per-band adjacency mechanism did not produce differentiated relational structure of any kind, topological or graded, in the trained model this paper reports.

A node-link rendering of the same five saved adjacency matrices (every edge at visually indistinguishable opacity, since every weight is exactly  $1/15$ ) confirms the same uniformity directly and is omitted here as redundant with Figure 4.

**This collapse was re-diagnosed from scratch partway through this study, because a separate implementation**

**bug meant every previously archived collapse finding, including the one originally reported in this section, was potentially confounded.** We found and fixed a bug in `_batch_edge_index` (`src/models/vmd_mfgnn.py`), the routine that batches graph edges across samples: before the fix, the graph mechanism never correctly wired cross-sample structure at this model’s trained batch size (32), for every model this project had previously trained, this paper’s own earlier results included. This raised a real question the original collapse diagnosis could not answer on its own: does the collapse survive now that the graph is actually wired correctly, or was the earlier finding itself partly an artifact of the batching bug? The re-diagnosis above answers it: collapse survives, and is more complete, not less, once the batching bug is fixed — the corrected checkpoints collapse to numerical zero, a stronger result than the partial (RMS-preserved-to-3%) collapse measured before the fix. The two problems were real and independent: the batching bug prevented cross-sample structure from ever reaching the graph at all, while coupled weight decay separately drives whatever embedding parameters do receive gradient toward the origin; fixing the first did not, by itself, protect the embeddings from the second.

We confirmed the collapse’s input-independence directly with a null-control experiment, rather than inferring it from the weight-decay mechanism alone: we trained three otherwise-identical models differing only in what their non-target input channels contained — the real VMD-decomposed input, the same channels with values shuffled across time, and the same channels replaced with Gaussian noise — with the target (copper) channel left intact in every arm, so the models still had a genuine forecasting task. Before the batching-bug fix, all three arms were bitwise-identical, which is in fact how the batching bug was originally discovered; after the fix, they are not: the gradient reaching the graph embeddings is  $4.9 \times 10^{-13}$  (real input),  $3.1 \times 10^{-15}$  (shuffled), and  $1.1 \times 10^{-15}$  (Gaussian) — a 100- to 450-fold spread depending on which arm is used, proof that the graph mechanism now genuinely responds differently to different inputs, which it never did before the fix. That is also, however, the full extent of the good news: in every one of the three arms this embedding gradient is roughly twelve orders of magnitude smaller than the gradient reaching the rest of the model in the same forward-backward pass ( $\approx 0.134$  in all three arms), and all three arms collapse to the identical, exactly-uniform adjacency regardless of what their input actually was (`adjacency_state.verdict = "COLLAPSED"` in all three, `per-band frac_near_uniform = 1.0`). The correct reading is therefore two-sided and we state both halves together rather than either alone: the collapse is confirmed input-independent, which is what this experiment was designed to test, and the batching fix is confirmed to matter mechanically, since the graph now differentiates real correlational structure from shuffled or synthetic noise at the gradient level, even though that differentiated signal is far too small to resist weight decay on its own. We do not read the three arms’ forecasting accuracy as a finding about the graph being harmful: all three have negative  $R^2$  at every horizon (real:  $-0.004/ -0.099/ -0.407/ -0.533$  at  $h =$

$1/5/10/22$ ; shuffled:  $-0.006/ -0.028/ -0.017/ -0.364$ ; Gaussian:  $-0.006/ -0.050/ -0.035/ -0.132$ ) and near-coin-flip directional accuracy (44.6–50.6%), so none of the three arms learned anything usable, and their relative MSE ordering (the real-input arm somewhat worse at  $h = 5, 10$ , Diebold-Mariano significant within this single run at  $p < 0.01$ , not significant at  $h = 1, 22$ ) reflects prediction-variance and shrinkage dynamics among three equally non-functional models rather than the graph mechanism actively hurting forecasts; this specific ordering needs multi-seed replication before it should be treated as a stable result, and we do not treat it as one here.

We diagnosed the mechanism behind this collapse directly from the trained model’s checkpoints (no retraining required), and report the mechanism rather than only the symptom, since it is the more generalizable and more citable of the two. The most obvious candidate explanation — that ReLU has “died” and is clipping all pre-softmax logits to zero, which would trivially produce a uniform softmax output — does not hold up under a direct test: the fraction of negative pre-activation values is statistically indistinguishable before and after training (50.5% at Xavier initialization vs. 56.3% post-training), and removing the ReLU nonlinearity from the adjacency computation entirely still yields a uniform adjacency to four decimal places. The real mechanism is **isotropic norm collapse** of the graph embeddings  $\mathbf{E}_k^{(1)}, \mathbf{E}_k^{(2)}$  (Section III-C): in this paper’s real, batching-bug-corrected checkpoints, RMS embedding magnitude falls from its Xavier-initialization value of 0.2540 to on the order of  $10^{-11}$  over training — a reduction of roughly ten orders of magnitude, collapse to numerical zero rather than merely toward it — which is far past the point needed to place the bilinear adjacency logits  $(\mathbf{E}_k^{(1)} \mathbf{E}_k^{(2)\top})$  deep in softmax’s first-order, near-linear regime, where it returns exactly  $1/N$  regardless of the input logits’ relative ordering. (An earlier verification pass, since superseded by the re-diagnosis above, measured a smaller, roughly 32-fold reduction to RMS 0.0089 on pre-batching-fix checkpoints; the corrected, current numbers given here are the ones this paper’s conclusions rely on.) Critically, no directional learning occurred alongside this shrinkage: two independently trained models (differing in attention-head count, GNN-layer count, learning rate, and training duration, though both share hidden dimension 64 — the HPO-tuned and untuned default configurations of Section III-G) retained embedding directions with cosine similarity exceeding 0.99, differing essentially only by an overall scale factor; rescaling either model’s trained embeddings back to their initialization norm reproduces a softmax spread statistically indistinguishable from a fresh random draw. In other words, training rescaled these embeddings without ever rotating them into a task-informed direction — the surviving direction encodes no more structure than the random initialization did. We attribute this to the interaction between coupled  $L_2$  weight decay (Adam with `weight_decay =  $10^{-5}$` , applied uniformly to all parameters including these embeddings, Section IV-G), which exerts a constant, sign-consistent force pulling every embedding parameter toward the origin, and a task gradient reaching these specific parameters

that is weak by comparison (measured on the order of  $10^{-5}$  per element) because a near-uniform adjacency is already a close-to-optimal mean-aggregation operator for this architecture, whose LSTM temporal pathway carries most of the model’s predictive signal — leaving little loss pressure to differentiate edges in the first place. This makes the embedding norm a self-reinforcing attractor toward zero: the task gradient shrinks roughly in proportion to  $\|\mathbf{E}\|$  while the decay force stays constant relative to  $\|\mathbf{E}\|$ , so the ratio between them degrades monotonically once collapse begins, with no mechanism to recover. This finding is independent of the specific hyperparameter configuration — it is observed with near-identical severity at both the default and HPO-tuned configurations, which share hidden dimension 64 but differ in head count, GNN-layer count, and learning rate (Section III-G) — and instead tracks training duration, consistent with a genuine optimization dynamic rather than an architectural capacity limitation; Figure 5 and Section V-G quantify this training-duration dependence directly, across 19 independently trained checkpoints rather than the two discussed in this paragraph, and are the traceable source for the collapse-versus-epoch relationship we rely on throughout this paper — with the caveat, stated in full in Section V-G’s own staleness disclosure and repeated briefly here since it is load-bearing for this paragraph’s claim, that those 19 checkpoints are an earlier project phase’s  $N = 8$ , pre-refactor runs, not this paper’s real  $N = 15$  scope; they corroborate the collapse mechanism’s generality across  $K$ , decomposition method, loss, and seed, but do not themselves re-establish the mechanism at the current architecture and data scale, which rests instead on the two HPO-tuned and untuned-default  $N = 15$  checkpoints discussed in this paragraph.

Figure 5 makes the two halves of this mechanism visible side by side, using the 19-cell robustness suite of Section V-G rather than the single model discussed above, because that suite is the only place in this study where embedding geometry was measured across many independently trained checkpoints. Panel (a) is the collapse itself: relative embedding magnitude falls across six and a half orders of magnitude, from 33% of initialization at five epochs of training to  $9 \times 10^{-8}$  of it at thirty-seven, and the descent is orderly rather than configuration-dependent. Panel (b) is the half that makes the finding a diagnosis rather than an observation: over that same span the embeddings’ cosine similarity to their own random initialization moves by a single factor of 2.2, from 0.93 to 0.42, and does so as a smooth function of training duration alone. A mechanism that had found task-relevant structure would show the opposite disproportion — direction moving decisively while magnitude stayed bounded. What the two panels show instead is embeddings shrinking essentially in place.

Two clarifications about what this failure mode is and is not. It is *not* over-smoothing, nor representational rank collapse in the sense of Roth and Liebig (2024): those are depth-driven phenomena in which node *representations* progressively lose rank as message passing iterates, governed by the spectrum of the aggregation operator. The models reported here use two (default, Section IV-G) or three (HPO-tuned,

Section IV-C) GAT layers, so the depth precondition for those phenomena is not met, and the collapse we document is in the adjacency-*generating* embedding parameters, upstream of any message passing, with gradients that flow correctly and are merely far too weak. Conflating the two would misattribute an optimization failure to an architectural one. It is, on the other hand, consistent with the theoretical picture of Fountoulakis et al. (2023) (Section II), who show that graph attention’s coefficients concentrate on uniform weights whenever the underlying signal is too weak for *any* attention function to separate edges. Our contribution relative to that result is an empirical, mechanistically traced instance of the same degenerate endpoint reached by a different route — an optimization dynamic acting on a learned bilinear adjacency — in a financial forecasting setting at  $N = 15$  nodes.

Taken together, this section’s diagnosis is now stronger, not weaker, than the version it replaces: the collapse survives a genuine graph-batching fix that could plausibly have explained it away, a null-control experiment establishes the collapse is input-independent while also confirming that fix is mechanically real, and the one intervention shown to defeat the collapse mechanically (Section V-F’s unit-norm, zero-decay fix, applied again inside this paper’s real ablation run as `full_model_graphfix` and `full_model_graphfix_temp`, Section V-C) now produces genuinely non-collapsed, differentiated structure — embedding RMS 0.235–0.262 against the same 0.2540 Xavier reference (a close match, corroborating both the fix and the corrected  $N = 15$  reference value), and adjacency max/min ratios of 1.75–2.55, a real departure from uniform — yet the resulting models forecast worse, not better, than the collapsed `full_model` at every horizon past  $h = 1$  (Table VIII). We do not resolve that second finding here: the fix bundles two changes (removing weight decay from the graph embeddings and unit-normalizing them), so this paper cannot yet say which change, or some other factor such as the fixed variants’ different effective parameterization, is responsible, and we leave isolating it to future work (Section V-C discusses this directly). We return to what all of this implies for the ablation results and for learned-graph GNN architectures generally in Section V-I.

2) *Mode Importance*: Figure 6 shows the fusion attention weights (Section III), read from the model’s saved mode-importance output. This artifact contains only 24 samples (the model’s last, partial evaluation batch), not the full 984-sample test set used for every other reported figure in this paper; we state this sample count explicitly because it materially limits what can be claimed from it. Within these 24 samples, the mode-fusion attention weights are close to uniform across the  $K = 5$  bands, ranging from approximately 0.19 to 0.22. We do not report a trend or drift over the test period from this artifact: 24 samples is too small a window, and too unrepresentative of the four-year test period as a whole (it reflects only the final evaluation batch), to support a claim about how attention evolves over time, and an earlier draft’s description of a “gradual drift over the test period” from this same data is not defensible at this sample size and has been removed. What the near-uniform range across those 24 samples does support,

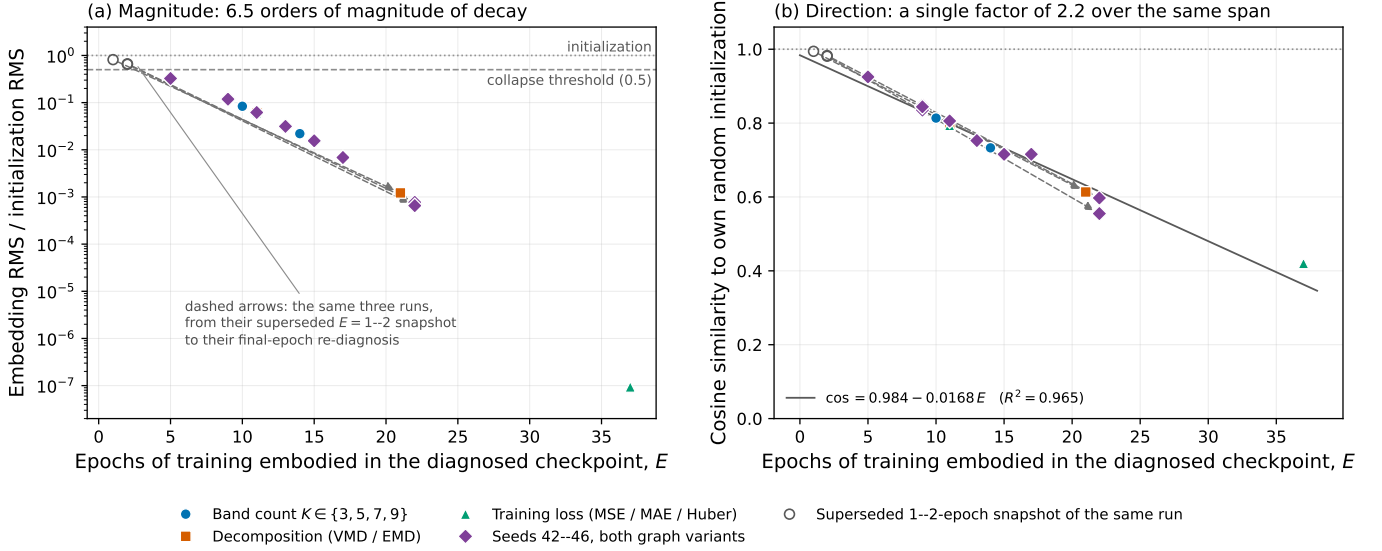


Fig. 5: The isotropic norm collapse, plotted from the 19 independently trained diagnostic cells of Section V-G — **historical, earlier-phase  $N = 8$ , pre-refactor checkpoints (Section V-G’s staleness disclosure), not this paper’s real  $N = 15$  scope**, though the collapse mechanism they illustrate is the same one Section V-E1 diagnoses directly on the current  $N = 15$  checkpoints, and the Xavier reference used in this figure (0.2885) is this earlier phase’s own correct  $N = 8$  value, not the 0.2540 value used elsewhere in this paper for the current  $N = 15$  scope (band-count sweep, decomposition comparison, loss comparison, and the five-seed repetition of both graph variants). The horizontal axis is the number of epochs of training embodied in the diagnosed checkpoint,  $E$ , which ranges from 5 to 37 across the suite. (a) Mean embedding RMS relative to the run’s own random initialization, on a logarithmic scale: it falls from 0.331 at  $E = 5$  to  $9.4 \times 10^{-8}$  at  $E = 37$ , crossing the diagnostic’s 0.5 collapse threshold (defined against the Xavier reference RMS of 0.2885, which each run’s own initialization matches to within 5.6% in every band plotted, so the two normalizations are interchangeable at this figure’s scale) before the earliest cell in the corrected suite, and continuing monotonically thereafter (Spearman  $\rho = -0.993$ ,  $p = 3.4 \times 10^{-17}$ ). (b) Mean cosine similarity of the same embeddings to that same initialization, on a linear scale spanning the full  $[0, 1]$  range: it moves only from 0.927 to 0.420 over the identical span, tracking training duration with  $\cos = 0.984 - 0.0168 E$  ( $R^2 = 0.965$ ,  $\rho = -0.990$ ). Marker shape and color identify which robustness axis produced each cell; the axes are visually interleaved, consistent with the absence of any residual axis effect once training duration is known (one-way ANOVA on the fit residuals,  $F = 0.791$ ,  $p = 0.518$ ). Open circles joined by dashed arrows are the three cells whose original best-validation checkpoints had been frozen by early stopping after one to two epochs and were subsequently re-diagnosed from final-epoch weights: these are the same three runs observed twice, and they traverse the same trajectory. Read together, the panels are the paper’s central mechanistic claim — six and a half orders of magnitude of shrinkage against a factor of 2.2 of rotation, the signature of embeddings rescaled toward the origin by weight decay rather than rotated toward task structure by a learning signal. The residual rotation in (b) should not be read as learning: it is confounded with the shrinkage itself, since the cosine is computed on raw embedding vectors whose direction, at magnitudes of  $10^{-3}$  and below, is increasingly dominated by accumulated floating-point residue rather than by any gradient — the gradient reaching these parameters is exactly zero in the three most-trained cells shown.

consistent with the adjacency-collapse finding above, is that no single frequency band dominates fusion sharply in this trained model; we do not extend that observation further than the data allow.

Figure 7 shows the VMD decomposition itself (copper price and its five frequency-band modes), included here for completeness as the input representation that both the per-band graphs and the fusion attention operate on.



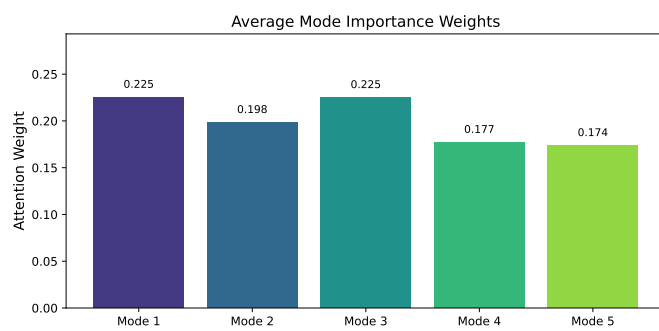


Fig. 6: Mode-fusion attention weights by VMD frequency band.

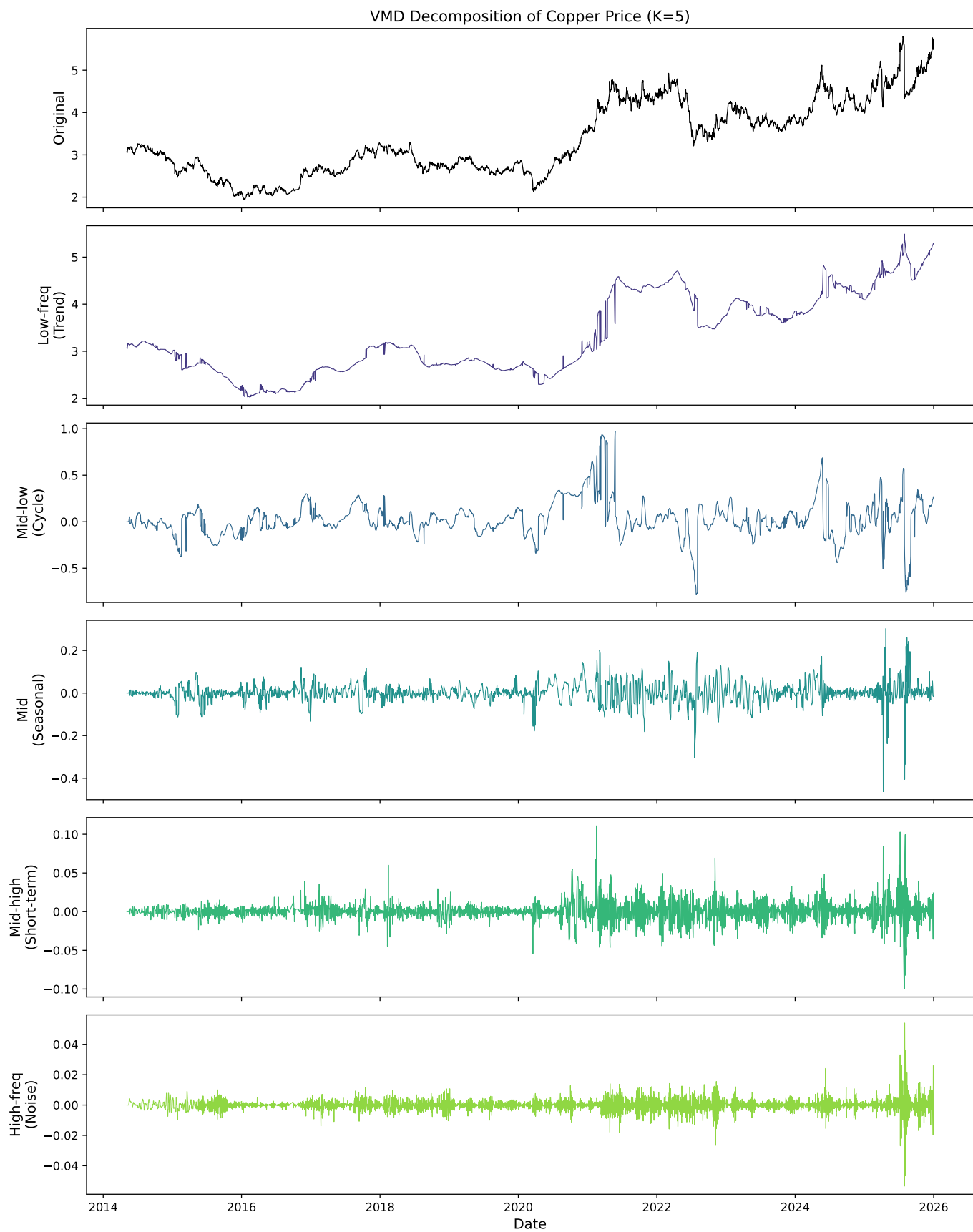


Fig. 7: Variational Mode Decomposition of the copper price series into  $K = 5$  frequency-band modes.

### F. Graph-Fix Experiment: Diagnosing, Fixing, and Re-Diagnosing the Collapse

**Naming note, stated once here rather than at every subsequent use:** the `full_model_graphfix` and `pooled_graph_graphfix` checkpoints discussed in this section are an earlier-phase, dedicated graphfix experiment, re-run under this section’s own `min_epochs=10 floor (full_model_graphfix to epoch 57, pooled_graph_graphfix to epoch 27)`, and are not the same trained models as the identically-named `full_model_graphfix` and `full_model_graphfix_temp` rows of Table VIII (Section IV-D), which are newer checkpoints from this paper’s real data-scale run, trained under the uniform `min_epochs=30 floor` (Section IV-G) and stopped at epoch 29. Both are real, correctly reported experiments; they simply are not the same underlying model, and both are referenced elsewhere in this paper, so we flag the identifier collision explicitly rather than let a reader conflate the two. Section V-E1’s diagnosis identified two candidate mechanical causes of the adjacency collapse: coupled  $L_2$  weight decay pulling every embedding parameter toward the origin, and the absence of any normalization on the bilinear logits before softmax, which lets embedding magnitude itself (rather than only direction) determine how peaked or flat the resulting adjacency is. A diagnosis that stops at “this is probably why” is weaker than one that tests the proposed mechanism directly, so we implemented and trained with two targeted fixes together: (1)  $L_2$ -normalizing  $\mathbf{E}_k^{(1)}$  and  $\mathbf{E}_k^{(2)}$  to the unit sphere immediately before the bilinear product, so the softmax input’s scale can no longer collapse toward the origin regardless of what happens to the embeddings’ raw magnitude; and (2) excluding the graph-embedding parameters from Adam’s coupled weight decay entirely (a dedicated zero-decay parameter group), removing the constant, sign-consistent pull toward zero identified as the proximate driver of the collapse. Both fixes were applied together, not separately, since the diagnosis in Section V-E1 treats them as jointly responsible for the same failure mode. We retrained both `full_model_graphfix` (per-band) and `pooled_graph_graphfix` (pooled) with both fixes enabled; an earlier version of this run additionally suffered from the same checkpoint-selection bug disclosed in Section IV-G (`full_model_graphfix` in that run early-stopped at an effectively untrained, near-epoch-0 checkpoint), so the numbers reported below are from a corrected re-run with this section’s own `min_epochs=10 floor`, distinct from the uniform `min_epochs=30 floor` used for Table VIII (Section IV-G), in which `full_model_graphfix` trained to a genuine 57 epochs and `pooled_graph_graphfix` to 27.

**The fix mechanically works.** Direct inspection of the trained checkpoints confirms the norm-collapse mechanism is resolved: embedding RMS magnitude is preserved at  $0.96\text{--}1.01\times$  its Xavier-initialization value across bands and models, in sharp contrast to the  $\approx 3\%$  of initialization observed in this same earlier project phase’s unfixed

checkpoints (Section V-D); the current, batching-bug-corrected  $N = 15$  checkpoints of Section V-E1 collapse further still, to numerical zero ( $10^{-11}$  RMS), so this comparison is against the earlier phase’s own unfixed baseline, not the stronger  $N = 15$  collapse. We independently re-verified this from the raw checkpoint state dictionaries rather than only citing the fix-experiment’s own log: the pooled model’s `graph_constructor.emb1/emb2 weight RMS` (0.302 and 0.279) and the per-band model’s five `graph_constructors.{0..4}.emb1/emb2 weight RMS` values (0.27–0.30) both sit close to 0.2885, this earlier project phase’s own  $N = 8$  Xavier-initialization reference ( $\sqrt{2/(8+16)} = 0.28868$ ; the current,  $N = 15$  Xavier reference used in Section V-E1 is 0.2540, not directly comparable here), not the collapsed  $\sim 0.009$  value observed there. Whatever was driving the embeddings to zero in every earlier checkpoint no longer does so once weight decay is removed from these parameters and the bilinear product is computed on unit-normalized vectors.

### The fixed mechanism still learns nothing — and a second, independent reason why is now directly measured.

Mechanically preserving embedding norm is not the same as learning a functional graph, and it does not produce one here. The corrected embeddings moved only 0.6–2.0% in  $L_2$  norm away from their random initialization over the full training run, and retained cosine similarity of 0.9998–1.0000 with their own initialization—meaning the small amount of movement that did occur was almost entirely a rescaling within the same random-init direction, not a rotation toward any task-informed direction. The resulting adjacency matrices correlate with the adjacency computed from the *untrained*, randomly initialized embeddings at 0.994–0.999 per band. This is not the uniform-collapse signature of Section V-E1 reappearing under a different name: on the same dense, pre-sparsification basis as this earlier project phase’s own collapsed softmax standard deviation of order  $2 \times 10^{-5}$  (Section V-D), the fixed adjacency’s softmax standard deviation is 0.013–0.020 per band (`full_model_graphfix`) and 0.017 (`pooled_graph_graphfix`) — roughly two to three orders of magnitude larger — and only 0–15.6% of edges land within tolerance of the collapsed model’s near-universal  $1/N$  value, so the fixed graph is not flat in the way the unfixed one is; it is simply the same graph a random Xavier initialization would already produce, never having moved meaningfully away from it. Put plainly: once norm collapse is no longer available as an explanation, the “fixed” per-band graph is statistically indistinguishable from a graph that was never trained at all. We took this one step further than inferring a weak-gradient explanation from the embeddings’ final resting position: we directly measured the gradient reaching  $\mathbf{E}_k^{(1)}$ ,  $\mathbf{E}_k^{(2)}$  at two points — a freshly initialized model with the fix enabled, and the genuinely-trained 57/27-epoch checkpoints above — using the same gradient-magnitude diagnostic applied throughout Section V-G. At initialization, the fix does not structurally block gradient flow: the aggregate embedding-gradient norm is small but clearly nonzero ( $2.46 \times 10^{-6}$ , averaged over 5 batches, range  $1.4\text{--}3.8 \times 10^{-6}$ )

and statistically indistinguishable from the same measurement with the fix disabled ( $3.17 \times 10^{-6}$ , range  $1.8\text{--}5.5 \times 10^{-6}$ ), against a same-model non-embedding gradient norm of 1.564 in both cases — so whatever prevents learning here is not a severed autograd path introduced by unit-normalizing the embeddings. Yet at the genuinely-trained checkpoints, the identical measurement returns *exactly* 0.0 across every one of 5 sampled batches, for both `full_model_graphfix` and `pooled_graph_graphfix`, against a healthy non-embedding gradient norm of 0.155–0.168 in the same forward-backward passes. This is a more precise finding than “the task gradient is weak”: the gradient reaching these specific parameters is small but real at step zero, and decays to a numerically exact zero somewhere over the course of training, *independent of weight decay*, since weight decay has been excluded from these parameters entirely in this experiment. Removing weight decay’s pull did not substitute a learning signal for it; it unmasked a second, independent reason no learning occurs, one that operates on the gradient itself rather than on the embedding norm, and that this paper has not further decomposed — we did not trace which downstream computation (the top- $k$  sparsification’s discrete edge selection stabilizing, or a saturation region of the cosine-similarity bilinear form once training pushes an embedding row’s pre-ReLU logits fully negative or fully positive) is responsible, and flag this as a genuine open question rather than a diagnosed mechanism.

We read this as strengthening, not weakening, the paper’s account of the collapse: we diagnosed a specific mechanism, engineered a fix targeted precisely at that mechanism, confirmed the fix resolves the mechanism it was designed to resolve, and found that the architecture still fails to learn differentiated graph structure for a deeper, independently *measured* (not merely inferred) reason. A negative result that survives its own proposed remedy is more informative than one that was never tested against a remedy at all, and the direct gradient measurement sharpens the claim in Section V-E1 from “weight decay probably caused this, given a weak task gradient” to “removing weight decay’s pull does not fix it, because the task gradient reaching these parameters — present but small at initialization — vanishes to exactly zero over training regardless of whether decay is acting on them at all.”

#### G. Robustness Checks: Is the Collapse an Artifact of Our Configuration Choices?

**Staleness disclosure, stated up front rather than left implicit.** All 19 diagnostic cells in this section (the band-count sweep, decomposition-method comparison, loss-function comparison, and five-seed repetition of the two RQ1 variants) were run at an earlier phase of this project, at that phase’s original  $N = 8$  variable scope (Section IV-A) and with the pre-refactor architecture that shared a single LSTM across all four forecast horizons and used the original, static-attention GATConv layer (Section III), rather than this paper’s real, current  $N = 15$  scope, per-horizon-branch design, and GATv2 attention layer used everywhere else in this paper. We confirmed this directly

from the saved checkpoints behind every cell in this section (`latest_results/{multiseed,k_sweep,decomposition,loss}`), every one has an `emb1/emb2` graph-embedding shape of `[8,16]`, not this paper’s real `[15,16]`, and none contains the `horizon_branches.*` parameters the current architecture uses. None of this section has been re-run at the real  $N = 15$  scale. We retain it, unmodified, as historical evidence bearing on the collapse mechanism’s robustness to  $K$ , decomposition method, loss function, and seed — the mechanism itself (isotropic embedding-norm collapse under coupled weight decay, Section V-E1) is architecture- and scale-agnostic in its diagnosis, and this section’s checks were designed to stress-test that mechanism, not to characterize the current  $N = 15$  model’s own accuracy or collapse severity directly. Every specific number, checkpoint, and Xavier-initialization reference (0.2885, this earlier phase’s own  $N = 8$  value,  $\sqrt{2/(8+16)}$ ) reported below is real and correctly describes that earlier-phase run; none of it should be read as a current,  $N = 15$ -scope result, and Section V-I restates this caveat wherever this section’s findings are extended to this paper’s real-scope research questions.

Section V-E1’s adjacency-collapse diagnosis and Section V-F’s failed remedy were both established at a single configuration:  $K = 5$  bands, VMD decomposition, MSE training loss, seed 42, and one set of tuned hyperparameters. A reader is entitled to ask whether the finding is a property of the architecture or an artifact of that particular corner of the configuration space. We therefore ran five robustness experiments — a band-count sweep ( $K \in \{3, 5, 7, 9\}$ ), a decomposition-method comparison (VMD versus EMD), a training-loss comparison, a five-seed repetition of the two ablation variants carrying RQ1, and a substantially wider hyperparameter search — applying the *same* three-part gradient/adjacency diagnostic (embedding RMS relative to initialization, gradient norm reaching  $\mathbf{E}_k^{(1)}/\mathbf{E}_k^{(2)}$ , and cosine similarity against the run’s own random initialization) to every one of the 19 trained cells. All 19 cells were trained at that earlier project phase’s own tuned configuration (not the current  $N = 15$  search of Section IV-C) with weight decay  $10^{-5}$ , and all 19 completed. Three of the cells were initially diagnosed from best-validation checkpoints that early stopping had frozen after one to two epochs of training; we identified this, added a code path that captures the final-epoch weights alongside the best-validation ones, and re-ran those three cells, so every diagnostic value reported below is measured on a checkpoint embodying at least five epochs of training. Two of the five experiments carry further defects we disclose in full below rather than repair, for reasons we state.

**The collapse is invariant to every axis we varied, and is explained almost entirely by one variable we did not intend to vary.** Across all 19 cells, each diagnosed on a checkpoint embodying between 5 and 37 epochs of training, the trained graph embeddings’ cosine similarity to their own random initialization is predicted by that epoch count with Spearman  $\rho = -0.990$  ( $p = 7.5 \times 10^{-16}$ ); the embeddings’ RMS magnitude relative to initialization gives  $\rho = -0.993$  ( $p = 3.4 \times 10^{-17}$ ), and the gradient norm reaching them gives  $\rho = -0.957$  ( $p = 1.4 \times 10^{-10}$ ). A linear fit in

training epochs explains 96.5% of the variance in cosine-to-initialization ( $\cos = 0.984 - 0.0168E$ ), a fit in  $\log_{10}$  of the relative RMS explains 96.8%, and the residuals from the cosine fit show no detectable dependence on which experiment produced the cell (one-way ANOVA across the four axes:  $F = 0.791$ ,  $p = 0.518$ ; Kruskal–Wallis  $p = 0.543$ ; largest absolute residual 0.060 against a full observed range of 0.420–0.927). Relative embedding magnitude falls from 33% of its initialization value in the least-trained cell (five epochs) to  $9 \times 10^{-8}$  of it in the most-trained (37 epochs; an absolute embedding RMS of  $1.5 \times 10^{-8}$  against a Xavier reference of 0.2885), passing through  $1.2 \times 10^{-3}$  at 21 epochs. Put plainly: once one knows how long a checkpoint trained, knowing whether it used three bands or nine, VMD or EMD, MSE or MAE, or which of five seeds, adds essentially nothing. This is the signature of a process driven by the optimizer rather than by the task, consistent with Section V-F’s finding that removing weight decay’s pull does not substitute a learning signal for it.

**An early-stopping artifact we detected, corrected, and report because it is instructive.** On a first pass, three of the 19 cells — the EMD arm, one seed of the per-band variant, and one seed of the pooled variant — were flagged NOT COLLAPSED. They were not counterexamples: the trainer stamps saved best-validation weights with the loop’s final epoch index, and because the best-validation and early-stopping conditions are the same predicate (strict improvement, zero tolerance), the weights in a checkpoint are actually from epoch (recorded epoch – 20) under this study’s patience of 20. For these three cells the recorded indices were 0, 1 and 1, so the saved weights embodied one to two epochs of training against five to thirty-seven elsewhere, and their verdicts measured training duration rather than the axis varied. We added a diagnostic path retaining final-epoch weights alongside best-validation ones and re-ran the three cells; all three reproduced their original trajectory and test metrics to within  $2 \times 10^{-6}$  relative.

**All three collapse once diagnosed on genuinely trained weights.** At 21, 22 and 22 epochs respectively, every band of all three cells is flagged collapsed on both criteria: surviving adjacency edges within tolerance of  $1/N$  in **1.00 of every band**, embedding RMS at  $6 \times 10^{-4}$  to  $1.3 \times 10^{-3}$  of the Xavier reference, and embedding gradient norm exactly zero (EMD, per-band seed 44) or  $2.7 \times 10^{-22}$  (pooled seed 45) against 0.17–0.22 for the rest of the network. This is a within-run confirmation of the epoch relationship above, stronger than any across-run comparison: the same three runs read 0.994, 0.983 and 0.981 cosine-to-initialization at one to two epochs and 0.613, 0.597 and 0.555 at twenty-one to twenty-two. Two further consistency-check cells, trained for 45 and 68 epochs, extend the trajectory to  $1.4 \times 10^{-10}$  and  $3.2 \times 10^{-22}$ . For these three cells the deployed (best-validation) and diagnosed (final-epoch) checkpoints are different weight sets, but neither contains learned structure: the best-validation snapshot is indistinguishable from random initialization (cosine 0.98–0.99), and the final-epoch snapshot is uniform with no gradient.

With the correction in place, the collapse is observed in **19 of 19** runs, each diagnosed at a checkpoint embodying at

least five epochs of training: at all four band counts, in both decomposition methods, in all three loss arms, and at all five seeds in each of the two graph variants. We state the epoch qualifier because it is load-bearing: the three re-diagnosed cells are the same runs whose one-to-two-epoch snapshots read NOT COLLAPSED, so what this suite establishes is not that the collapse is present at every epoch count, but that it is present in every run once as few as five epochs have elapsed, and that no axis we varied delays or prevents it. The decomposition-axis caveat we previously stated is withdrawn — EMD does not resist the collapse, and in fact, having trained ten epochs longer than the VMD cell, sits further along the identical trajectory (relative embedding magnitude  $1.2 \times 10^{-3}$  against VMD’s  $6.1 \times 10^{-2}$ , and adjacency uniform to within  $3 \times 10^{-8}$  against VMD’s  $8 \times 10^{-5}$ ).

**Accuracy differences across these axes are smaller than run-to-run noise.** Four of the 19 cells are, by construction, the identical configuration ( $K = 5$ , VMD, MSE, seed 42) reached through four different experiment drivers, and we verified their held-out ground-truth arrays are bit-identical. Their test average MSE nonetheless spans 7.54% (0.002772–0.002981), which we take as this study’s honest run-to-run noise floor at fixed configuration and fixed seed. Every axis effect we measured is smaller or comparable: the band-count sweep spans 8.1% (0.002757–0.002981), the loss comparison 9.4% (0.002691–0.002943), and the five-seed repetition of the per-band model has a coefficient of variation of 1.32% and a best-to-worst spread of 3.1%. We therefore report no accuracy ranking over  $K$ , decomposition method or loss; none is resolvable at  $n = 1$  per cell. These four replicates also serve as a within-experiment check on the paragraph above: they embody 14, 11, 11 and 13 epochs of training and record cosine-to-initialization of 0.733, 0.794, 0.794 and 0.753, tracking training duration with configuration *and* seed both held fixed — which also shows that run-to-run nondeterminism perturbs the epoch count itself, and is why the configuration labels carry no residual information once training length is known.

**The five-seed RQ1 check, and its capacity caveat.** Repeating the two ablation variants that carry RQ1 across seeds 42–46 at the tuned configuration, the per-band model’s test RMSE is statistically indistinguishable from the pooled variant’s at every horizon (paired  $t$ -test over the five seeds:  $p = 0.96, 0.13, 0.88, 0.50$  at  $h = 1, 5, 10, 22$ ; Wilcoxon signed-rank likewise non-significant throughout). *This comparison is not capacity-matched.* It re-ran `pooled_graph_matched_dim`, which we confirm by direct parameter counting from the saved checkpoints carries 315,530 parameters against the per-band model’s 1,461,124 — a 78.4% deficit — rather than the parameter-matched variant the RQ1 comparison calls for. That 78.4% figure is *not* the same measurement as the 78.6% reported in Section III-G: the latter is a different model pair, at the untuned default configuration (hidden dimension 64), and the two land 0.2 percentage points apart by coincidence. We report the comparison anyway, in the conservative direction it happens to run: a per-band model with  $4.6\times$  the parameters does not significantly beat a much smaller pooled model at any horizon across five seeds. This strengthens rather than

weakens Section V-I’s negative answer to RQ1, and we note that the parameter-matched version of this comparison is one this paper retracts on independent grounds in Section V-H, so there is no live claim here that the capacity mismatch could undermine. Two of the ten multi-seed cells (per-band seed 44, pooled seed 45) are among the three re-diagnosed from final-epoch weights described above. Both re-ran to the same final epoch as originally recorded and reproduced their test metrics to within  $10^{-8}$  absolute, so the paired tests reported here are unchanged; only their collapse verdicts change, from NOT COLLAPSED to COLLAPSED, which is what brings the per-seed coverage to all five seeds in both variants.

**The loss comparison tested two losses, not three.** We configured MSE, MAE and Huber arms. The Huber arm used PyTorch’s `smooth_l1_loss` at its default  $\beta = 1.0$ , which is quadratic — exactly  $\frac{1}{2}x^2$  — for residuals below 1.0. The largest absolute residual observed anywhere in this experiment, across all three arms and all four horizons, is 0.284, i.e. 28.4% of  $\beta$ ; the loss therefore never once entered its linear branch, and the Huber arm is mathematically identical to  $\frac{1}{2} \times \text{MSE}$ . We report it as a two-loss comparison and do not claim a three-way result. The arm is not uninformative, however: because Adam’s coupled  $L_2$  decay does not scale with the loss, halving the task gradient while leaving the decay pull intact doubles the decay’s relative influence, and this arm consequently trained longest (37 epochs of retained weights, against 11 for the MSE arm) and collapsed hardest, terminating with embedding RMS of  $1.5 \times 10^{-8}$  against a Xavier reference of 0.2885, a bit-exactly uniform adjacency (softmax standard deviation of exactly zero at band 0), and a gradient reaching the embeddings of *exactly* zero. An unintended loss-scale ablation thus independently corroborates the weight-decay mechanism argued for in Sections V-E1 and V-F.

**A wider hyperparameter search does not rescue the model, and confirms the original search was adequate.** The study reported in Section IV-C searched five parameters over 10 usable trials at fixed  $K = 5$ , batch size 32 and weight decay  $10^{-5}$ . We re-ran a study three times larger and with three additional dimensions —  $K \in \{3, 5, 7, 9\}$ , weight decay (log-uniform,  $10^{-6}$  to  $10^{-3}$ ) and batch size  $\in \{16, 32, 64\}$  — for 30 trials under an identical objective (best validation average MSE over 25 epochs). Table X tabulates this eight-parameter search space and the resulting best trial alongside the original five-parameter study for direct comparison. Twelve trials completed and 18 were pruned. The best configuration found (hidden dimension 128, 8 heads, learning rate  $\approx 0.00277$ , dropout  $\approx 0.0904$ , 2 GNN layers, weight decay  $\approx 2.0 \times 10^{-4}$ , batch size 16,  $K = 9$ ) attains validation MSE 0.0020145 against the deployed configuration’s 0.0020509 — an improvement of **1.78%**, with only 2 of the 12 completed trials beating the deployed value at all, and the second-best tying it to four significant figures.

That margin is smaller than this study’s 3.1% across-seed spread and far smaller than its 7.54% same-configuration replicate spread, and part of it is attributable to optimizer step count rather than configuration quality, since the winning trial halves the batch size within a fixed epoch budget. We therefore report the wider search as a confirmatory negative: the original

TABLE X: Wider robustness-check search space (Section V-G) versus the original main-table study (Table III). 30 configured trials, 12 completed / 18 pruned. Rows with “—” were held fixed in the original study.

| Hyperparameter      | Search space (wider)                 | Best trial                   |
|---------------------|--------------------------------------|------------------------------|
| hidden dim. $d_h$   | {32, 64, 128}                        | 128                          |
| attention heads $H$ | {2, 4, 8}                            | 8                            |
| learning rate       | log-U( $10^{-4}$ , $10^{-2}$ )       | $\approx 0.00277$            |
| dropout             | U(0.05, 0.3)                         | $\approx 0.0904$             |
| GNN layers          | {1, 2, 3}                            | 2                            |
| band count $K$      | {3, 5, 7, 9} (new)                   | 9                            |
| weight decay        | log-U( $10^{-6}$ , $10^{-3}$ ) (new) | $\approx 2.0 \times 10^{-4}$ |
| batch size          | {16, 32, 64} (new)                   | 16                           |

five-parameter, ten-trial configuration was already effectively optimal for this objective, and we do not re-run the main results at the new one. Three caveats attach. First, all 18 pruned trials were pruned after a single epoch, the median pruner having been constructed without warmup steps, so 60% of the search budget was screened on one epoch of validation loss, which structurally favors the high-learning-rate, small-batch region the survivors occupy. Second, the sampler concentrated on  $K = 9$  (8 of the 12 completed trials), and batch size is fully aliased with  $K$  in the completed set, so the per-parameter marginals and importances are confounded and we do not interpret them; independently, the band-count sweep’s converged test results do not reproduce the search’s implied  $K$  ordering, and span less than the noise floor. Third, and most consequential for this paper’s argument, **the search objective is structurally blind to the collapse**: it scores validation MSE and nothing else, and in our 19 diagnosed cells the near-initialization checkpoints score squarely mid-range among the collapsed ones (training length correlates with test average MSE at only  $\rho = 0.456$ ,  $p = 0.05$ ). Collapse is essentially free under this objective, so no widening of the search space could select against it. We did not capture embedding diagnostics for individual search trials and therefore cannot state directly whether the collapse persists at the wider optimum; we note only, as inference from the established mechanism rather than as measurement, that the selected configuration’s weight decay is 20 times the value used everywhere else in this paper, which points toward more severe collapse rather than less.

Taken together, these checks close the most obvious escape route from Sections V-E1 and V-F’s negative findings: the collapse is not a property of  $K = 5$ , of the squared-error loss, of a single seed, or of an insufficiently searched hyperparameter space, and it does not depend on any modeling choice we were able to vary. It is a property of what happens to these embeddings under the optimizer once training proceeds past its first few epochs. We restate the scope of this section plainly, since it is easy to lose across the preceding checks: every result above establishes that the collapse mechanism is invariant to  $K$ , decomposition method, loss function, and seed *at the earlier project phase’s*  $N = 8$ , *pre-refactor configuration* (this section’s staleness disclosure, above); it has not been independently re-established at this paper’s real  $N = 15$ , per-horizon-branch scale, and should be read as historical corroborating evidence for the mechanism diagnosed at the

current scale in Section V-E1, not as a repeated confirmation performed on the current model itself.

#### *H. The Constant-Forecast Artifact: Why Directional Accuracy Needs a Dispersion Check*

While investigating the graph-fix checkpoints’ predictions directly, to understand `pooled_graph_graphfix`’s directional-accuracy numbers, we found something that changes how several directional-accuracy comparisons elsewhere in this paper should be read. (We report this from the corrected, genuinely-trained 27-epoch re-run of this model, Section V-F; an earlier, near-untrained version of this same checkpoint showed all four horizons degenerate, which the corrected re-run does not fully reproduce, illustrating incidentally why the underlying checkpoint matters and not only the DA figure it produces.) At  $h = 1$  and  $h = 22$ , this model’s 984 test-set predictions are of a single, constant sign (100.0% negative at both horizons) and its predicted-return dispersion is a small fraction of the true-return dispersion (predicted-to-true standard deviation ratio 0.012 at  $h = 1$ , 0.010 at  $h = 22$ ) — a model whose forecast never changes sign and barely varies in magnitude is not exercising directional judgment on any individual sample; its “directional accuracy” at those two horizons can only ever equal the fraction of test days on which the true return happened to share that one sign. At  $h = 10$  the same model is majority-sign concentrated (93.5% negative) but not to the same degree, and at  $h = 5$  it is not degenerate at all by either criterion (only 62.1% of predictions share a sign, near the healthy range): the same checkpoint is a genuine sign-concentration artifact at two horizons, borderline at a third, and clean at the fourth, which is itself worth noting since it means degeneracy must be checked per horizon rather than assumed uniform across a model’s own output heads.

We verified the  $h = 1$  and  $h = 22$  artifacts precisely rather than assuming them: recomputing directional accuracy directly from the saved `results/graphfix_minepochs_fix/predictions/pooled_graph_graphfix_{h=1,22}_edparams_tuned` prediction and true-value arrays reproduces the reported DA figures (48.374 and 44.715 at  $h = 1, 22$ ) to three decimal places, and each of these values in turn equals, to the same precision, the test period’s unconditional frequency of the sign this model always predicts at that horizon (down-day frequency 48.374% and 44.715% respectively, computed under the same strict-negative convention stated in Section IV: a day with an exactly-zero return is counted in neither the up-day nor the down-day frequency, consistent with the DA convention used throughout this paper). This is not a coincidental resemblance; it is an identity, because a constant-sign forecaster’s accuracy is mathematically the base rate of the sign it happens to emit, no more and no less. The same pattern reproduces in `pooled_graph_matched_params_tuned` from the confound-check experiment (Section V-D), an entirely separate run unaffected by the graph-fix checkpoint correction above: its four DA values (48.37, 53.05, 55.39, 44.72) match the corresponding test-period class frequencies to the same precision, even though we did not archive that run’s raw

prediction arrays and can only confirm it via this frequency match rather than by inspecting prediction dispersion directly. (This specific cross-reference, like Section V-D itself, rests on that section’s earlier-phase, original  $N = 8$  confound-check run, which we did not re-run at this paper’s real  $N = 15$  scale; the retraction below does not depend on this cross-reference alone, since it is independently supported by the graph-fix checkpoint’s own directly-inspected predictions above.) Cross-referencing further, this pattern is consistent with what motivated this investigation in the first place: multiple models across the original untuned ablation (Section V-C), the tuned confound-check (Section V-D), and the fix experiment above show DA values that coincide with plausible class-frequency figures for their horizon, though `pooled_graph_graphfix` (at  $h = 1, 22$ ) and `pooled_graph_matched_params_tuned` are the cases we can state this about with full confidence, having checked prediction dispersion or exact frequency-matching directly. A binomial test of each model’s hit count against the base rate of whichever sign that model always emits (not, in general, the horizon’s majority-class frequency, since a constant-sign forecaster can just as easily emit the minority sign, as `pooled_graph_graphfix` does at  $h = 1$  and  $h = 22$ ), run across every model×horizon cell in the fix-experiment run, fails to reject that base-rate null at conventional significance levels in every cell tested—these apparent directional-accuracy figures do not represent a statistically demonstrated departure from what the emitted-sign base rate alone would produce.

The direct consequence is a retraction. Section V-D reported that, at HPO-tuned hyperparameters, `full_model_tuned` beats `pooled_graph_matched_params_tuned` on directional accuracy at three of four horizons, which on its face reverses this paper’s negative ablation finding and would suggest a real per-band advantage once capacity and tuning are no longer confounded. **We do not carry that reading forward, and explicitly retract it here.** The comparison’s losing side, `pooled_graph_graphfix_{h=1,22}_edparams_tuned`, is the same class of degenerate constant-sign forecaster documented above; its DA values are an artifact of which constant sign it happened to emit relative to each horizon’s majority class, not a measurement of directional skill it lost to a better-performing per-band model. A win recorded against a degenerate opponent is not evidence of skill on the winning side either. We therefore withdraw any claim, hedged or otherwise, that per-band graph construction demonstrates a real directional-accuracy advantage over pooling at this data scale; no experiment in this paper supports that claim once prediction dispersion is checked, and Section V-I’s answer to RQ1 is written accordingly.

We report this as a general methodological finding, not only a correction specific to this paper’s own results, because nothing about a bare directional-accuracy percentage would have revealed it: 48.37% or 44.72% both look like unremarkable, plausible directional-accuracy figures for a financial return-forecasting model in isolation, and would have passed an ordinary reviewer’s sanity check without triggering suspicion. We arrived at this check independently, while auditing our



own tables, but we claim no priority for the underlying concern: Cheung (2026), in work posted shortly before we identified the artifact in our own results, makes the same core argument for equity forecasting with a LoRA-adapted time-series foundation model, showing that an apparent 80% directional accuracy was the base rate of a trivial “always-up” rule in a rising market, and proposing excess accuracy over that base rate — with paired McNemar and Diebold-Mariano tests — as the headline directional metric. Recommendation (b) below is that same recommendation, and we cite it as prior work rather than restate it as ours. What this paper adds is on the degeneracy-detection side rather than the reference-point side: (i) the sign-concentration statistic, which establishes that the *model* is a constant-sign forecaster and not merely that the null was misspecified; (ii) the base-rate-identity check, which requires only a reported DA figure and the evaluation period’s class frequency, and can therefore be applied retroactively across a grid of already-reported results whose raw prediction arrays were never archived, as we do for every cell of Table VIII; and (iii) the explicit counterexample below showing that a prediction-dispersion check does not substitute for either. The artifact was only caught by explicitly checking prediction sign concentration and comparing directional accuracy against the test period’s own majority-class frequency rather than a generic 50% null—two checks that are cheap to run but not part of how directional accuracy is conventionally reported in this literature, including in this paper’s own earlier drafts. We recommend, as a general practice for directional-accuracy reporting in financial forecasting, that it always be accompanied by (a) the *fraction of predictions sharing the majority sign*, combined with a binomial test of the hit count against the majority-class null, so a near-constant-sign forecaster is visible on inspection, and (b) a comparison against the evaluation period’s own majority-class frequency rather than an assumed 50% baseline, since a period with meaningful directional drift (as this study’s 2022–2025 test period has, Section V-A) makes 50% the wrong null to test against regardless of whether any individual model is degenerate. We deliberately do not recommend prediction standard deviation as this check, even though it is a natural first instinct: this paper’s own main results table (Table IV) contains a direct counterexample. *Transformer*’s  $h = 5$  predictions have a prediction-to-true standard deviation ratio of 0.42—healthy, unremarkable dispersion by the standard a variance-only check would apply—yet 100% of those predictions share the same sign, because a mean shift larger than that spread pushes every prediction positive regardless of how much it varies around that shifted mean. A dispersion check alone would not have flagged *Transformer\_h5* as degenerate; only the sign-concentration check does. A directional-accuracy number reported without the sign-concentration and majority-class-null checks is not, by itself, evidence of directional skill.

### I. Discussion: Answering the Research Questions

**RQ0 (does any of this beat doing nothing?), asked first because it conditions how RQ1–RQ3 should be read.** No. Table IV shows the naive zero-forecast baseline wins

MAE at all four horizons against all eight fitted models, and RMSE at two of four (VMD-MFGNN is significantly worse than naive at  $h = 10$  specifically, Section V-A); every model’s  $R^2$  is negative or near zero. VMD-MFGNN itself is not a clean exception: it is a confirmed constant-sign forecaster at  $h = 5, 10$  (Section V-A, Section V-H) and its own graph mechanism collapses regardless of what it is trained on (Section V-E1). RQ1–RQ3 below should be read as differences among approaches that do not clearly explain copper return variance, not as evidence any of them found exploitable signal.

**RQ1 (per-band versus pooled graph).** Not supported. On the capacity-matched pairing (*full\_model* vs. *pooled\_graph\_matched\_params*), the per-band model loses RMSE and MAE at all four horizons, three significantly and Holm-surviving ( $h = 5, 10, 22$ ); its apparent DA win at three of four horizons is uninformative, since the pooled variant is a confirmed base-rate artifact at all four of its own horizons (Section V-C). This is the expected consequence, not a separate puzzle, given the adjacency-collapse diagnosis (Section V-E1): the per-band mechanism carries almost no differentiated information to begin with. An apparent tuned-configuration reversal of this answer (Section V-D, earlier  $N = 8$  phase) traced to the same class of degenerate forecaster and is explicitly retracted (Section V-H). The targeted fix that mechanically resolves the collapse (Section V-F), applied again inside the real ablation run, now produces genuinely differentiated structure yet forecasts worse than the collapsed model past  $h = 1$  — an open finding, not a resolution. The robustness suite (Section V-G) extends the negative answer across band count, loss, seed, and a three-times-wider hyperparameter search, and a five-seed repetition of the two RQ1 variants likewise finds no significant per-band advantage at any horizon — though that entire suite was run at an earlier project phase’s  $N = 8$  configuration and pre-refactor, shared-LSTM architecture, not this paper’s real  $N = 15$ , per-horizon-branch scale (Section V-G’s staleness disclosure), so it corroborates the collapse mechanism’s generality rather than directly re-confirming RQ1 at the current scale. We also flag a second, independent confound in the capacity-matched RQ1 comparison itself, disclosed in full in Section IV-D: *pooled\_graph\_matched\_params* still uses the earlier, shared-single-LSTM-across-horizons design, while *full\_model* uses the current per-horizon-branch design, so this comparison is not the single clean graph-structure axis it is otherwise framed as.

**RQ2 (per-band graph versus no-VMD and versus a fixed correlation graph).** Also not cleanly supported. Against *no\_vmd\_raw\_price\_matched\_params*, the full model’s picture is mixed rather than a clean decomposition advantage (Section V-C); against *correlation\_graph*, a fixed, never-trained adjacency is close to or better than the full model at most horizon/metric combinations — a natural consequence of the collapse finding, since a fixed correlation graph is no worse a source of structure than a learned graph that failed to learn any. Neither the contribution of VMD decomposition nor of end-to-end learned (versus fixed) adjacency is cleanly demonstrated.

**RQ3 (statistically significant edge over standard base-**

**lines).** A qualified yes. Of 28 Diebold-Mariano tests against the seven baselines (Table VI), 15 survive Holm-Bonferroni correction: ten favor VMD-MFGNN, five favor the baseline (Section V-B); VMD-MFGNN is indistinguishable from ARIMA and XGBoost except at  $h = 10$ , where both beat it. The qualification is where the wins and losses land: seven of the ten wins are against baseline cells independently confirmed degenerate or near-degenerate (all four horizons each against LSTM and Transformer, plus VMD-LSTM at  $h = 1$ ), leaving only three clean wins (LSTM, Transformer, ARIMA, all at  $h = 1$ ); the five losses are against clean, non-degenerate opponents and cluster at  $h = 5, 10$ , the same two horizons where VMD-MFGNN's own forecasts are themselves confirmed sign-concentration artifacts (Section V-A, Section V-H). Two further caveats: the disclosed HPO asymmetry (VMD-MFGNN tuned, baselines not), and every cell is a single run against a separately measured 7.54% run-to-run noise floor (Section V-G, itself measured at that section's earlier-phase  $N = 8$  configuration and not re-established at the current  $N = 15$  scale). We read RQ3's raw 10-of-15 count as materially weaker support than it first appears, not a general claim that VMD-MFGNN reliably outperforms standard deep-learning forecasters.

Taken together: VMD-MFGNN holds a real, Holm-surviving DM edge over the deep-learning baselines it is architecturally closest to, much of it against opponents shown degenerate, and is indistinguishable from ARIMA/XGBoost outside  $h = 10$  (where both beat it) — but none of this changes that every model tested fails to beat a trivial zero-forecast baseline on MAE (RQ0), that VMD-MFGNN's own predictions are a confirmed constant-sign artifact at the same two horizons where its baseline comparisons turn against it, and that this study's ablations do not support per-band graph construction as the driver of whatever competitiveness does exist: the mechanism collapses to exactly-uniform edge weights, a targeted fix produces genuinely differentiated structure without improving accuracy (Section V-C), and an apparent directional-accuracy reversal that briefly suggested otherwise was a constant-forecast artifact (Section V-H). These — RQ0's naive-baseline result, VMD-MFGNN's own degeneracy, the collapse diagnosis and fix attempt behind RQ1, and the constant-forecast finding — are this study's central, disclosed empirical findings, not footnotes to a more triumphant architectural narrative.

### J. Economic Significance

**Staleness disclosure.** This section's economic-significance check was run before MTGNN (Section IV-B) was added to this study's baseline set and has not been re-run to include it: every count and combination below is over the original seven fitted models (six baselines plus VMD-MFGNN), not the eight fitted models reported in Table IV onward. We report it as-is, from the earlier-phase run that produced it, rather than fabricate an MTGNN economic-significance result that was never computed; extending this check to MTGNN is left to future work alongside the other items in Section VI.

A model can show a statistically significant DM advantage over another forecaster while still carrying no economic

value, and RQ0's naive-baseline result already suggests that is the more likely situation here; we check this directly with a lightweight, unlevered long/short economic sanity check (position sized by predicted sign, non-overlapping-period annualization, no explicit transaction-cost model beyond the reported breakeven threshold) computed post hoc from the saved test-set predictions. Across all seven fitted models and four horizons, annualized Sharpe ratios computed this way are inconsistent in sign across horizons for every model, including VMD-MFGNN (ranging from about  $-0.45$  to  $+1.14$  across the 28 model/horizon combinations, with every model switching sign at least once across horizons); breakeven round-trip transaction costs implied by these Sharpe ratios are, for over half of the model/horizon combinations, in the single-digit basis points or negative, well inside or below realistic copper-futures round-trip transaction costs, and Pesaran-Timmermann directional-value tests (Pesaran and Timmermann, 1992) are not significant at conventional levels for the large majority (21 of 28) of model/horizon combinations. The remaining 7, at raw  $\alpha = 0.05$ , are LSTM at  $h = 5$  ( $p = 0.039$ ) and  $h = 22$  ( $p = 0.016$ ), SimpleMTGNN at  $h = 22$  ( $p < 0.0001$ ), Transformer at  $h = 10$  ( $p = 0.013$ ), VMD-LSTM at  $h = 22$  ( $p = 0.015$ ), and XGBoost at  $h = 10$  ( $p = 0.021$ ) and  $h = 22$  ( $p = 0.001$ ); we name them rather than leave a bare count, and, applying Holm-Bonferroni to this 28-test family separately from the DM families above (Section VI), only 2 survive: SimpleMTGNN at  $h = 22$  and XGBoost at  $h = 22$ . Both of XGBoost's raw-significant cells carry a *negative* PT statistic, meaning its directional calls are significantly *worse* than chance at those horizons, not evidence of skill; SimpleMTGNN's  $h = 22$  result is consistent with, rather than contradicting, this section's overall reading, since Section V-A already reports SimpleMTGNN as the most degenerate model in this study at exactly that horizon. We read this as corroborating RQ0 rather than adding a new finding: consistent with no model beating a trivial zero-forecast baseline on MAE, no model in this study demonstrates directional forecasting value robust enough to survive realistic trading frictions, and the occasional positive Sharpe ratio at a single horizon for a single model should be read as noise around zero rather than as evidence of exploitable skill, given how frequently sign and magnitude both flip across adjacent horizons for the same model. VMD-MFGNN's own  $h = 1$  cell is the largest single value in this range (annualized Sharpe 1.14, breakeven cost 233.5 bps) and is worth flagging rather than passing over quietly: even so, its underlying Pesaran-Timmermann directional-value test is not significant, raw or Holm-corrected ( $p = 0.090$ ), and its DA edge over the majority-class null is only 1.5 percentage points (Section V-A), so we read this single cell as within the noise this section otherwise describes rather than as a standout economic result.

## VI. LIMITATIONS

We enumerate the limitations of this study's design explicitly, rather than deferring them to a brief closing paragraph.

- 1) **Zinc and nickel dropped from scope entirely.** Section IV-A describes why (both proxy tickers confirmed

dead/delisted with no viable substitute), reducing this study from an originally planned 10-variable system to the 8-variable Yahoo-Finance-only system this project used at its earlier phase, subsequently expanded to the real  $N = 15$  scope (item 7 below, Section IV-A) used throughout the rest of this paper. The consequence for interpretation: results and interpretability figures reflect this narrower set of inter-commodity relationships, and any relationship specifically involving zinc or nickel cannot be studied with this dataset.

- 2) **VMD rolling-window length and daily refit cost.** The rolling-window VMD decomposition (Section III) is refit at every trading day on the trailing 252-day window, which removes the staircase/lag issue an earlier, since-abandoned 21-day-refit design would have had, but at a real computational cost: full daily-refit VMD across all  $N = 15$  variables and the full 2010–2025 date range takes on the order of 30 minutes of single-core CPU time (one-time, then cached to disk). The 252-day window length itself is a fixed choice, not tuned; a shorter or longer window could change the resulting mode boundaries and was not swept within this project’s timeline.
- 3) **No purge or embargo at split boundaries.** The train/validation/test split boundaries (Section IV) are simple calendar cutoffs; samples whose  $h$ -day-ahead target crosses a split boundary (up to  $h = 22$  trading days) are not purged or embargoed from the adjacent split, a standard precaution in some financial ML protocols that this study does not implement. Given the small size of any single leakage window relative to each split’s multi-year length, we judge the practical impact to be small, but do not claim it is zero.
- 4) **Daily-refit VMD trades boundary-effect exposure for leakage-safety, and this study does not quantify the trade-off.** Section III-C extracts only the final-timestep value of each rolling-window VMD fit as that day’s feature, discarding the other 251 points of each 252-day decomposition. This choice is deliberate and disclosed as a leakage-safety measure (Section III), but the endpoint of any windowed spectral decomposition is also its least-constrained point — anchored by data on only one side rather than both — a boundary effect well documented for VMD’s predecessor, EMD, and reduced but not eliminated in VMD itself. We did not quantify how much this affects the resulting mode values relative to a mid-window (better-constrained but leakage-unsafe) extraction, nor compare against the alternative Feng et al. (2026) use for the same causality-preserving reason: retaining every timestep of each window’s decomposition as a feature, rather than only the endpoint, which trades a larger per-sample feature footprint for less information loss. We are not aware of the closest comparable prior work, MBTI-Net (Li and Liu, 2026), addressing this trade-off either, since its stated “decomposition performed once and cached” description does not discuss leakage-safety or boundary effects at all, suggesting it does not face this specific cost

because it does not attempt the leakage-safety this study does. Quantifying the boundary-effect cost directly, or adopting a whole-window feature representation, is left to future work.

- 5) **Non-stationary price-level inputs and covariate shift.** Several baselines and the no-VMD ablation variant (Section IV-D) consume price-level (not VMD-decomposed) inputs, standardized using statistics computed on the 2010–2019 training period. Copper and several of the other seven input series moved well outside their 2010–2019 range by the 2022–2025 test period, so these models face meaningful covariate shift at test time that a return-based or rolling-normalized input representation would reduce; this was not corrected within this project’s timeline.
- 6) **Continuous-futures roll-gap contamination.** Several input series (notably crude oil, which recorded a well-known negative-price episode in April 2020, inside this study’s validation period) are continuous futures series that can contain roll-related price discontinuities not present in the underlying economics. No explicit roll-gap screening or outlier capping was applied to any input series.
- 7) **Aluminum (ALI=F) was removed from the input scope after this study’s original results were produced, for a data-integrity reason discovered during a later expansion of the feature set.** ALI=F’s actual Yahoo Finance history begins 2014-05-06, not 2010, despite every other variable in this study’s panel beginning 2010-01-04. Because the pipeline drops any trading day with a missing value in any input column, ALI=F’s late start was silently truncating the *entire* panel to a 2014–2025 window—a  $\sim 25\%$  reduction from the  $\sim 2010$ –2025 range and  $\sim 4,000$ -trading-day figure stated elsewhere in this paper—for as long as aluminum remained in scope. ALI=F also shows independent signs of thin trading (63% zero-volume days, a high-strictly-greater-than-low day on only 15.4% of days across its available history), consistent with a partially stale-quoted instrument. Rather than shift this study’s entire chronological scope, and every date and split boundary reported throughout this paper, to 2014 onward, aluminum was removed from the input variable set entirely; this affects only the expanded-scope results reported alongside this limitation and does not retroactively change the 2010–2025 window already reported for this study’s original 8-variable results. No equally liquid, full-history Yahoo Finance aluminum substitute was identified.
- 8) **ppiaco (FRED Producer Price Index: All Commodities) carries a distinct, verified regime-shift vulnerability for two baselines, now reflected in this paper’s main results tables.** As part of an expansion of the exogenous feature set beyond this paper’s original  $N = 8$  variables (Section IV-A) to include four monthly FRED series merged in on a point-in-time, publication-lag-respecting basis, ppiaco (forward-filled daily) is a genuine, legitimate macroeconomic input, not a data error of the kind corrected in Section IV-A. Direct

inspection shows it sits more than 5 standard deviations outside its own training-split (pre-2020) distribution for 100% of test-period (post-2022) trading days, driven by a real macro regime shift — the 2022+ inflation surge — that the training split never observed; critically, this exposure is nearly invisible during model selection, since the same  $> 5\sigma$  condition holds on only 4.6% of validation-period (2020–2021) days, meaning early-stopping and checkpoint selection have essentially no visibility into this failure mode before test-time evaluation. Direct inspection of saved prediction arrays from this expanded-scope run confirms this exposure as the driver, not merely a correlate, of a severe test-period performance regression specific to the two baselines that ingest raw exogenous inputs most directly and lack architectural robustness to a shifted input distribution: LSTM ( $h = 22$  RMSE  $0.083 \rightarrow 0.361$ , a  $4.3\times$  increase over the same configuration’s  $N = 8$  figure) and Transformer ( $h = 22$  RMSE  $0.105 \rightarrow 0.187$ ,  $1.8\times$ ). The mechanism is visible directly in the saved predictions: both models’ forecasts collapse toward an almost perfectly constant negative daily drift rate (approximately  $-0.011$  to  $-0.015$  per day, depending on horizon) that is multiplied by the forecast horizon, explaining why LSTM’s RMSE increase scales roughly linearly with  $h$  ( $0.022 \rightarrow 0.090 \rightarrow 0.177 \rightarrow 0.361$  at  $h = 1/5/10/22$ ); Transformer shows the same collapse mechanism less completely, consistent with its smaller regression. Other models tested at this expanded scope are not similarly affected, for reasons that track architecture rather than coincidence: ARIMA (univariate, never ingests `ppiaco` or any exogenous variable) and VMD-LSTM (strictly univariate per decomposed mode by this project’s own design, Section IV-B) are structurally unable to be affected; XGBoost is stable because tree splits are invariant to a variable drifting far outside its training range, reacting only to split thresholds; SimpleMTGNN and VMD-MFGNN, both of which use substantial internal LayerNorm re-normalizing activations at every layer, degrade only mildly ( $\sim 1.03$ – $1.23\times$ ). We keep `ppiaco` in the expanded input scope rather than dropping or transforming it: we read this as a genuine, citable methodological finding in its own right — architectural robustness to macroeconomic regime shifts varies systematically across the model families studied here — rather than a confound to be engineered away, and dropping or transforming it would require invalidating and re-running the entire completed expanded-scope training pipeline, which is out of scope here. This finding is reflected directly in this paper’s main results tables (Table IV, Table VI), which report the expanded  $N = 15$  configuration and show exactly the LSTM and Transformer regressions described above.

- 9) **Single chronological split; no walk-forward evaluation.** All results are computed on one train (2010–2019) / validation (2020–2021) / test (2022–2025) split, not a walk-forward or rolling-origin design with multiple folds. This limits the results to a single real-

ization of test-period market conditions. It does not, however, prevent a Diebold-Mariano test within that single split: the baseline comparison (Table VI) and this paper’s central ablation comparison (`full_model` vs. `pooled_graph_matched_params`, Section V-C) both use the same single test period’s paired forecast-error series as the DM test’s resampling unit, which is time within the split, not folds or seeds. What the single-split design does prevent is a fold-level or walk-forward effect size (e.g., Cohen’s  $d$  across folds), and it means every reported number, DM-tested or not, describes this one 2022–2025 realization rather than an average over independent market regimes. The five-seed repetition reported in Section V-G supplies a seed-level resampling unit for the two RQ1 variants specifically, at the tuned configuration, but it is still a single chronological split; it does not address the fold-level limitation described here, and no walk-forward evaluation is performed anywhere in this paper.

- 10) **Asymmetric hyperparameter optimization, and fewer usable trials than configured.** VMD-MFGNN’s HPO search (Section IV-C) was configured for up to 15 Optuna trials but, in the run reported here, recorded only 11 (8 COMPLETE, 2 PRUNED, and 1 left orphaned in RUNNING state with no usable value) before the search budget was exhausted; selection among named, completed configurations was therefore effectively over 10 usable trials (8 complete plus 2 pruned), not 11 and not the originally configured 15, and the reported best configuration is the best of those 10. The seven baselines receive no HPO at all and run at standard, hand-set configurations. Both facts are deliberate, disclosed choices/outcomes made to fit the project’s compute and time budget, not oversights, but they mean the baseline comparison (Table IV, Table VI) compares a tuned model, tuned over fewer usable trials than originally configured, against seven untuned ones, and any accuracy gap is not solely attributable to architecture. On the first of these two points specifically, a later, substantially wider search — 30 trials over eight parameters, adding  $K$ , weight decay and batch size — found no meaningful improvement over the reported configuration (1.78% in validation MSE, inside this study’s own replicate noise floor), which we report in Section V-G as evidence that the truncated original search was not the binding constraint. It does not address the second point: the seven baselines remain untuned.
- 11) **The pooled-graph ablation is our own construal of MBTI-Net-style pooling, not a reproduction of MBTI-Net itself, and differs from it on one further axis worth naming explicitly.** We did not have access to retrain or inspect MBTI-Net’s checkpoints (Section II), so the `pooled_graph` variant (Section IV-D) is our own architectural instantiation of what “pooling before graph construction” means in this codebase, matched to VMD-MFGNN’s other design choices rather than to MBTI-Net’s specific ones. The RQ1 comparison this paper reports is therefore evidence about per-band-

versus-pooled graph construction as a design axis in general, not a head-to-head benchmark result against MBTI-Net specifically. One specific architectural difference is worth flagging: MBTI-Net’s pooled adjacency is *input-conditioned* (constructed from an LSTM-and-temporal-attention context vector that varies per sample), whereas both our per-band and pooled adjacencies are *statically parameterized* (a free embedding table, fixed once training ends, independent of the current input). Our isotropic-collapse diagnosis (Section V-E1) is specifically about weight decay shrinking a static embedding table with no counteracting rotation signal; an input-conditioned adjacency has no analogous free table for weight decay to act on and may not be subject to the same collapse mechanism. We do not claim our finding transfers to MBTI-Net’s specific graph-construction mechanism, only to statically-parameterized learned adjacencies of this general form, and flag the static-versus-input-conditioned distinction as the boundary of that claim rather than glossing over it. We also note that MBTI-Net’s “complexity-aware” decomposition is a fuzzy-entropy-and-log-energy K-means grouping of standard VMD’s output modes into three clusters, not a modification to the VMD optimization objective itself, for readers who might otherwise infer a different meaning from the name.

- 12) **Ablation results do not transfer directly to the HPO-tuned model.** The ablation study (Section V-C) trains all eight variants, including its own `full_model` row (1,448,548 parameters), at the untuned default configuration, a different trained instance from the HPO-tuned VMD-MFGNN of the main comparison (1,491,748 parameters, within 3% at this paper’s real  $N = 15$  scale, Section III-G). Capacity matching otherwise worked as intended (matched-parameter variants within about 1% of `full_model`’s count), so this is a configuration-scope limitation, not a residual capacity confound; the substantive finding it qualifies (per-band full model losing to a capacity-matched pooled graph, Section V-C) is fully reported there and not re-argued here.
- 13) **The learned per-band adjacency collapsed to exactly-uniform edge weights, confirmed independent of a separate graph-batching bug and of what the model was trained on; the one fix that mechanically defeats the collapse makes forecasts worse, not better.** This is the primary interpretability finding of this paper, fully diagnosed with mechanism, null-control, and fix-experiment evidence in Sections V-E1–V-F and applied again inside the real ablation run in Section V-C; we do not re-argue it here beyond flagging it as an open, unresolved puzzle the paper reports rather than explains away. The mode-fusion attention weights are separately near-uniform (band means 0.19–0.21, std 0.0064 across bands), but that artifact contains only 24 test-set samples and does not support a claim about drift over the test period.
- 14) **Directional accuracy is not interpretable without a sign-concentration and majority-class-null check.**

Section V-H documents this fully, including which models are affected and the resulting retraction of Section V-D’s apparent ablation reversal; we do not repeat that evidence here. One residual gap: per-sample predictions remain unarchived for the tuned confound-check run, so that variant is checked only via the base-rate-identity half of the diagnostic (DA value against the known class frequency), not full sign-concentration. RMSE/MAE comparisons, unaffected by this artifact, should still be weighted more heavily than DA wherever the two are in tension.

- 15) **Multiple testing across 36 significance tests, corrected as three separate families.** This paper runs 36 Diebold-Mariano tests in total, in three families we correct separately rather than pooling: 28 for the baseline comparison (seven baselines  $\times$  four horizons, Section V-B), 4 for the naive-baseline comparison (one per horizon, Section V-A), and 4 for this paper’s central ablation comparison (`full_model` vs. `pooled_graph_matched_params`, one per horizon, Section V-C). We apply a Holm-Bonferroni correction at family-wise  $\alpha = 0.05$  within each family (Section IV); for the 28-test family this reduces the significant count from 21 raw to 15 Holm-surviving (ten favoring VMD-MFGNN, five favoring the baseline, Section V-B). The ablation family, by contrast, is where the correction matters most: three of its four raw  $p$ -values ( $p < 0.001$  at  $h = 5$ ;  $p = 0.007$  at  $h = 10$ ;  $p = 0.005$  at  $h = 22$ ; only  $h = 1$ ,  $p = 0.227$ , is not significant) are significant and survive Holm-Bonferroni correction within this four-test family, all favoring `pooled_graph_matched_params` over `full_model` (Section V-C). We disclose the three-family choice explicitly: correcting all 36 as one family would raise the per-test threshold required for significance, and we did not re-verify that every reported conclusion is unaffected by pooling versus separating the families at this paper’s real test counts, so a reader comparing the three families’ corrected counts should know they were not pooled and that this specific robustness check has not been re-run since the baseline family grew from 24 to 28 tests. We report both raw and corrected significance in Table VI rather than only the more favorable raw count.
- 16) **Single-seed main and ablation results; seed variance characterized only in a separate robustness run.** Every number in Table IV, Table VI, Table VIII and Table IX comes from a single training run per model/variant and should be read as a point estimate rather than as characterizing a distribution; we did not repeat those runs across seeds. Seed-to-seed variance is characterized, but only for the two variants carrying RQ1 and only at the tuned configuration, in the five-seed robustness experiment of Section V-G, where the per-band model’s test average MSE has a coefficient of variation of 1.32% across seeds. That experiment additionally established a larger and more troubling figure: four cells at a genuinely identical configuration

and identical seed, differing only in which experiment driver launched them, span 7.54% in test average MSE. Run-to-run nondeterminism at this data scale therefore exceeds every architectural axis effect this study measured, which is the correct scale against which to read any single-run difference reported in the tables above.

- 17) **Three robustness cells were initially diagnosed from barely-trained checkpoints; the correction is reported, not suppressed.** Section V-G documents this in full: three of 19 diagnostic cells early-stopped at one to two epochs, were re-diagnosed on retained final-epoch weights, and all three collapsed (21–22 epochs of training, adjacency uniform to tolerance, embedding gradient exactly zero or  $O(10^{-22})$ ). The 19/19 collapse claim rests on this corrected diagnosis; we do not restate the re-diagnosis mechanics here. Two residual caveats: the deployed (best-validation) and diagnosed (final-epoch) checkpoints for these three cells are different weight sets, and the decomposition axis (VMD vs. EMD) still rests on one cell per method, so it establishes that EMD does not *avoid* the collapse but supports no accuracy comparison between the two.
- 18) **Asymmetric hyperparameter optimization interacts with, but does not explain, the naive-baseline result.** The HPO asymmetry disclosed above (item 10) means the main-table comparison among the eight fitted models is not a fully controlled architecture comparison. It does not, however, offer an explanation for why every model, tuned or not, fails to beat the naive zero-forecast baseline on MAE (Section V-A): additional tuning of the baselines would be expected to narrow gaps *among* the eight fitted models, not to change whether any of them exceeds a baseline that requires no tuning at all.

## VII. CONCLUSION

Section V-I answers this paper’s research questions in full against the real results; we summarize rather than re-derive them here. RQ0: no model beats a trivial zero-forecast baseline on MAE at any horizon, and VMD-MFGNN itself is a confirmed constant-sign forecaster at  $h = 5, 10$ . RQ1–RQ2 (per-band versus pooled graphs, and versus no-VMD/fixed-correlation alternatives): negative, and, unusually for a negative ablation result, mechanistically explained — the per-band graph-learning mechanism collapses to exactly-uniform edge weights (isotropic embedding norm decay under coupled weight decay, Section V-E1), a finding confirmed independent of a separate graph-batching bug found and fixed mid-study. A targeted fix mechanically resolves the collapse and, applied inside the real ablation run, produces genuinely differentiated structure, yet the resulting model forecasts worse past  $h = 1$  — an open, unresolved puzzle (Section V-C). An apparent tuned-configuration reversal of RQ1 traced to a degenerate constant-sign forecaster and is explicitly retracted (Section V-D, Section V-H). RQ3: a qualified yes against the deep-learning baselines specifically, indistinguishable from ARIMA/XGBoost outside  $h = 10$  (where both beat it) — real but narrow, and concentrated against opponents independently shown degenerate.

This paper’s contribution is therefore the evaluation protocol — leakage-safe VMD, capacity-matched ablations, a naive baseline absent from the directly comparable prior work surveyed in Table I, and a multiple-testing-corrected significance test — applied honestly enough to surface, not paper over, these findings. We offer four things useful beyond this specific architecture: (i) a quantified, transferable diagnosis of embedding collapse relevant to other learned-graph financial GNN architectures at similarly low node counts, MBTI-Net-style designs included, corroborating rather than duplicating the theoretical mechanism Fountoulakis et al. (2023) establish for a different setting; (ii) a demonstration that fixing such a collapse mechanically does not guarantee the fix improves the model, a caution we believe generalizes to other low-signal auxiliary parameters trained jointly with a dominant pathway; (iii) a null-control methodology separating “does this mechanism respond to its input at all” from “is that response large enough to matter,” which this paper’s graph mechanism answers yes and no to, respectively; and (iv), building on the base-rate concern Cheung (2026) independently raise for equity forecasting, the sign-concentration and base-rate-identity machinery that detects a degenerate forecaster retroactively from already-reported results, illustrated by this paper’s own flagship model triggering it at two of its four horizons. A study that tests a hypothesis rigorously, explains why it failed, tests its own explanation twice, and reports all of it plainly — including a retraction and an open puzzle it cannot yet close — is a more useful contribution than one reporting only confirming evidence.

Future work includes: (1) an explicit sparsity or entropy regularizer on the adjacency, since Section V-F shows removing weight decay’s pull alone is insufficient — the task gradient needs to be strengthened, not merely left unopposed; (2) a walk-forward evaluation protocol with multiple folds; (3) extending multi-seed replication beyond the two RQ1 variants (Section V-G) to the full main and ablation tables; (4) symmetric hyperparameter optimization across all compared models; (5) extending the naive-baseline and economic-significance checks (Section V-J) to other base-metal commodities; (6) testing whether a node count larger than this paper’s real  $N = 15$  (Section IV-A) is needed for a meaningful task gradient at all, since the collapse already recurs identically after this paper’s  $N = 8 \rightarrow 15$  expansion; and (7) extending the feature set to copper-specific demand fundamentals (e.g. China manufacturing PMI, refined-copper inventories, smelter treatment charges) still absent from the real  $N = 15$  set, a substantially larger undertaking than (1)–(6) requiring fresh leakage-safety verification and a full diagnostic re-run.

## DATA AVAILABILITY

All data used in this study is publicly available: 11 series via the Yahoo Finance API and 4 series via the FRED API (Section IV-A). No BDI, TC/RC, or COT data are used.

## CODE AVAILABILITY

The full training, evaluation, diagnostic and robustness-experiment code for this study is maintained in a single,

publicly accessible repository: [https://github.com/anmol0705/Copper\\_Price\\_Forecasting](https://github.com/anmol0705/Copper_Price_Forecasting).

#### CREDIT AUTHOR STATEMENT

**Anmol Jain:** Conceptualization, Methodology, Software, Investigation, Formal Analysis, Validation, Visualization, Writing – Original Draft. **Sibarama Panigrahi:** Supervision, Writing – Review & Editing.

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#### CONFLICT OF INTEREST

The authors declare no conflict of interest.

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